

Latent Structure in Financial Statements: PCA, Factor Analysis, and Sector-Level Inference on 10-K Data

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Abstract

We explore the methodology and application of classical multivariate techniques, principal component analysis (PCA), hierarchical clustering, exploratory and confirmatory factor analysis (EFA/CFA), and multivariate inference (MANOVA, Hotelling's T^2), for uncovering latent structure in firm-level financial statements. After outlining the models and assumptions behind each method, we fit them to a panel of SEC 10-K financial metrics for 3,786 U.S. companies sourced from Financial Modeling Prep. PCA reveals that eight components capture roughly 65% of the variance, and EFA recovers three strong latent constructs, operational scale, balance sheet position, and profitability, plus a weak fourth residual. A confirmatory model, however, fits poorly (RMSEA ≈ 0.39); MANOVA shows that sector membership drives significant differences across most financial metrics, while the joint profiles of subsequently rising and falling stocks are statistically indistinguishable. Our analysis showcases the limitations of latent-variable methods on heavy-tailed, highly collinear accounting data. We then conduct a simulation study to display the intended use-case of factor analysis and PCA, and to empirically demonstrate how heavy tails and noise dimensions degrade recovery of a known latent structure.

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1 Introduction

Many statistical learning methods are designed to answer prediction questions: given a set of (many) co-variates, what outcome should we expect? Multivariate analysis asks a different question. Rather than predicting a response, it asks whether a high-dimensional set of measurements can be compressed into a small number of interpretable dimensions, and whether groups of observations differ systematically along those dimensions.

Annual 10-K filings provide a natural setting for this question. Each filing reports hundreds of financial metrics: revenue, margins, cash flows, balance sheet items, growth rates, and valuation ratios. These metrics are thoroughly analyzed by investors, yet they are also massively redundant. Gross profit is a linear function of revenue and cost of goods sold; EBITDA margin is a ratio of two other reported quantities; per-share figures scale mechanically with shares outstanding. The interesting question is whether, underneath this redundancy, a small number of latent constructs (how big a firm is, how stable its balance sheet is, how profitable its operations are) generate the observed metrics, and whether those constructs relate to subsequent stock performance.

A naive approach examines each metric separately, comparing (say) mean profit margin across sectors one variable at a time. This is interpretable but statistically fragile: with 200+ correlated metrics, univariate tests inflate false discoveries and ignore the joint structure that makes financial statements coherent. Multivariate methods address this directly. PCA and factor analysis model the correlation structure itself, clustering searches for firm “archetypes” in the reduced space, and MANOVA and Hotelling’s T^2 test group differences on all metrics simultaneously while accounting for their covariance.

The biggest challenge is that financial data violate the textbook conditions under which these methods behave well. Accounting variables are heavy-tailed, mechanically collinear, and contaminated by reporting artifacts (missing values, exact zeros, and occasional 1,000% price moves driven by micro-cap illiquidity). This project therefore has two aims. First, we apply the full multivariate toolkit to the 10-K panel and interpret what it finds. Second, we treat the messy results themselves as data: through a limitation analysis and a pair of simulation studies, we characterize when these methods succeed and how the specific pathologies of financial data cause them to fail.

2 Methods

2.1 The Problem

Let $\mathbf{x}_i \in \mathbb{R}^p$ collect the p financial metrics for firm i , with sample covariance matrix $\mathbf{S}$. All four methods below are ways of interrogating $\mathbf{S}$: PCA asks for the directions of maximal variance, factor analysis asks whether a low-rank-plus-diagonal decomposition explains the correlations, clustering asks whether firms concentrate in regions of the reduced space, and MANOVA asks whether group mean vectors differ once the covariance is accounted for.

2.2 Principal Component Analysis

PCA finds orthogonal linear combinations of the standardized variables that sequentially maximize variance. Writing the spectral decomposition $\mathbf{S} = \sum_{j=1}^p \lambda_j \mathbf{e}_j \mathbf{e}_j^\top$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$, the j -th principal component of firm i is $z_{ij} = \mathbf{e}_j^\top \mathbf{x}_i$, and the proportion of variance explained by the first k components is $\sum_{j \leq k} \lambda_j / \sum_{j \leq p} \lambda_j$ (Johnson and Wichern, ch. 8). Because PCA operates on variances, standardization matters: revenue is measured in billions of dollars while margins live on $[-1, 1]$, so we work with the correlation matrix throughout.

Two properties of PCA matter for this application. First, PCA is purely a variance decomposition. It makes no distinction between signal variance and noise variance, so heavy-tailed noise, which is ubiquitous in financial ratios, can rotate the leading eigenvectors away from the true structure. Second, components are defined only up to interpretability: nothing guarantees that a direction of high variance corresponds to an economically meaningful construct. Factor analysis addresses the second concern by positing a generative model.

2.3 Hierarchical Clustering

Given a distance matrix on the firms (we use Euclidean distance in the space of retained principal component scores), agglomerative clustering repeatedly merges the two closest clusters. The definition of “closest” is the linkage. Ward’s method merges the pair whose union minimizes the increase in within-cluster sum of squares, which tends to produce compact, similarly-sized clusters. Minimax linkage (Bien and Tibshirani, 2011) instead assigns each cluster a prototype, an actual firm, and defines the cluster radius as the maximum distance from the prototype to any member. This yields directly interpretable clusters: each group of firms comes with a representative company one can look up. Interpretability of the prototypes was our main motivation for including minimax alongside Ward.

Clustering here plays a descriptive role. It asks whether, in the reduced PC space, firms form sector-independent archetypes, and it inherits all the fragility of the PCA step that precedes it.

2.4 Factor Analysis

2.4.1 The Orthogonal Factor Model

Exploratory factor analysis posits that the standardized metrics are generated by $m \ll p$ common factors:

$$\mathbf{x}_i = \mathbf{\Lambda} \mathbf{f}_i + \boldsymbol{\varepsilon}_i, \quad \mathbf{f}_i \sim (\mathbf{0}, \mathbf{I}_m), \quad \boldsymbol{\varepsilon}_i \sim (\mathbf{0}, \boldsymbol{\Psi}),$$

with $\boldsymbol{\Psi}$ diagonal and $\mathbf{f}_i \perp \boldsymbol{\varepsilon}_i$. The implied covariance is $\boldsymbol{\Sigma} = \mathbf{\Lambda} \mathbf{\Lambda}^\top + \boldsymbol{\Psi}$: a low-rank component capturing shared variation plus variable-specific uniquenesses. Unlike PCA, the factor model separates common variance from idiosyncratic variance, which is exactly the distinction we want for financial metrics, where each line item mixes firm-level fundamentals with account-specific noise. The loadings $\mathbf{\Lambda}$ are identified only up to rotation; we use varimax to push each variable toward loading on a single factor, which aids interpretation.

The maximum likelihood fit assumes multivariate normality. Financial metrics are decidedly non-normal, so we treat MLE-based factorial output as a descriptive decomposition and reserve judgment on its standard errors. The simulation study in Section 4 quantifies how much this matters.

2.4.2 Confirmatory Factor Analysis

CFA reverses the workflow: the analyst specifies which variables load on which factors, fixing all other loadings to zero, and the model is fit and evaluated as a hypothesis. We assess fit with three standard indices. The comparative fit index (CFI) compares the model to an independence baseline, with values above roughly 0.95 indicating good fit. RMSEA measures misfit per degree of freedom, with values below roughly 0.06 desirable. SRMR is the standardized average residual correlation, with values below 0.08 desirable (Hu and Bentler, 1999). A CFA that fails these thresholds tells us the constraints we imposed, and hence the theory behind them, are inconsistent with the observed covariance.

2.5 MANOVA and Hotelling’s T^2

To test whether financial profiles differ across groups, we use multivariate analysis of variance. For g groups with mean vectors $\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_g$, MANOVA tests $H_0 : \boldsymbol{\mu}_1 = \dots = \boldsymbol{\mu}_g$ via Wilks’ lambda,

$$\Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{W} + \mathbf{B}|},$$

the ratio of within-group to total generalized variance. Small values reject H_0 . For two groups, the analogous statistic is Hotelling’s T^2 ,

$$T^2 = \frac{n_1 n_2}{n_1 + n_2} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)^\top \mathbf{S}_p^{-1} (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2),$$

with $\mathbf{S}_p$ the pooled covariance. Both procedures assume multivariate normality within groups and homogeneous covariance matrices. With sample sizes in the thousands the tests are reasonably robust to non-normality, but the covariance-homogeneity assumption deserves skepticism across sectors as different as utilities and biotech, and we flag this in the limitation analysis.

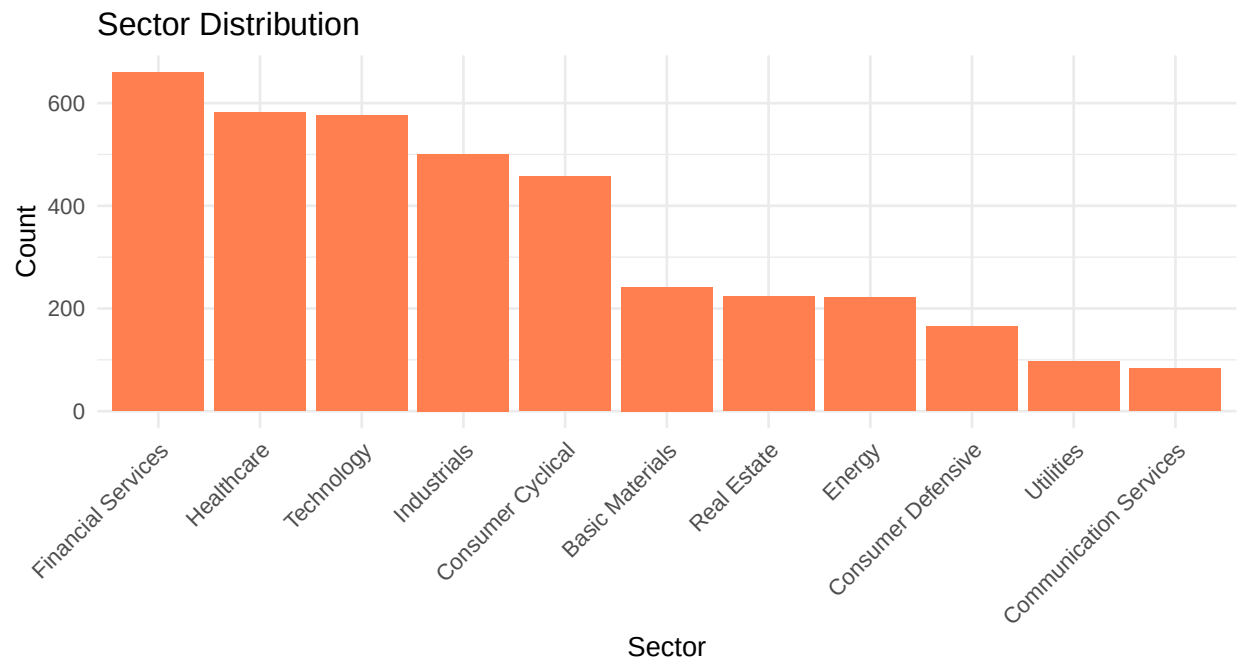
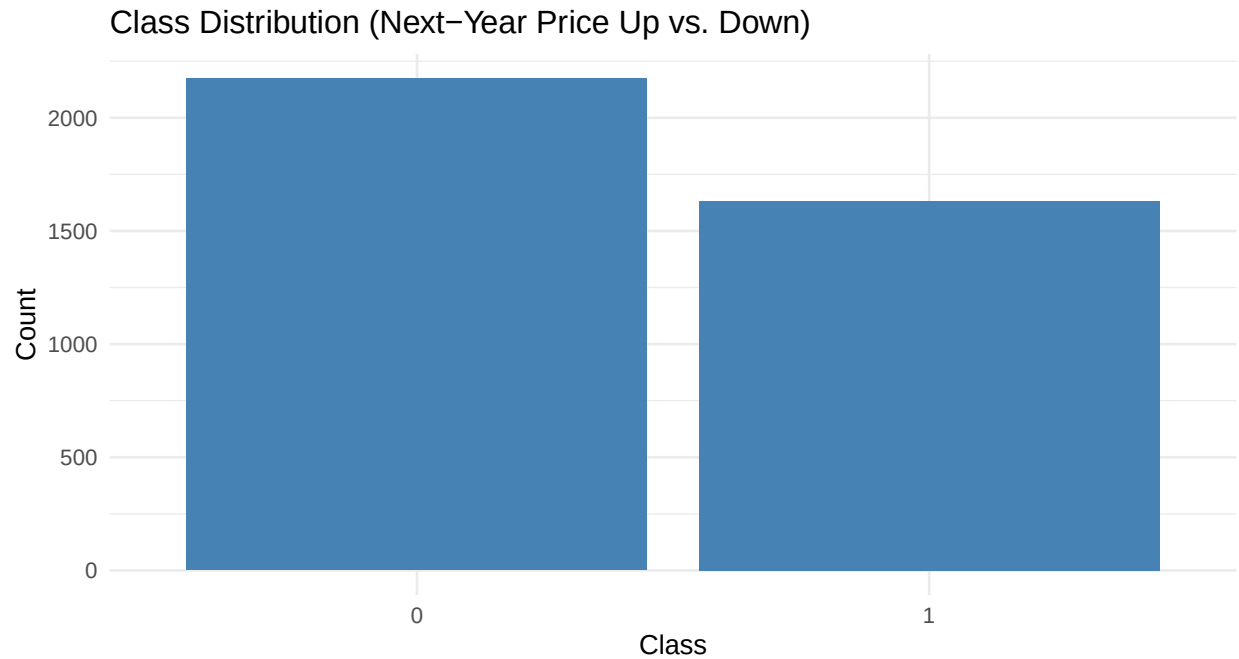
3 Data Analysis

3.1 Initial QA

The data come from Financial Modeling Prep and cover annual 10-K metrics for 3,786 U.S. companies across 11 sectors, spanning fiscal years 2014 through 2018.¹ Each yearly file contains roughly 225 financial metrics and ratios per firm, plus the sector label and two targets: the percent change in stock price over the following calendar year, and a binary up/down indicator (**Class**). We center the main analysis on the 2014 filing year, so the forward return is the 2015 price variation.

[1] 3808 225

¹Available at <https://site.financialmodelingprep.com/>. The five annual files are stored in **data/** in the project repository.



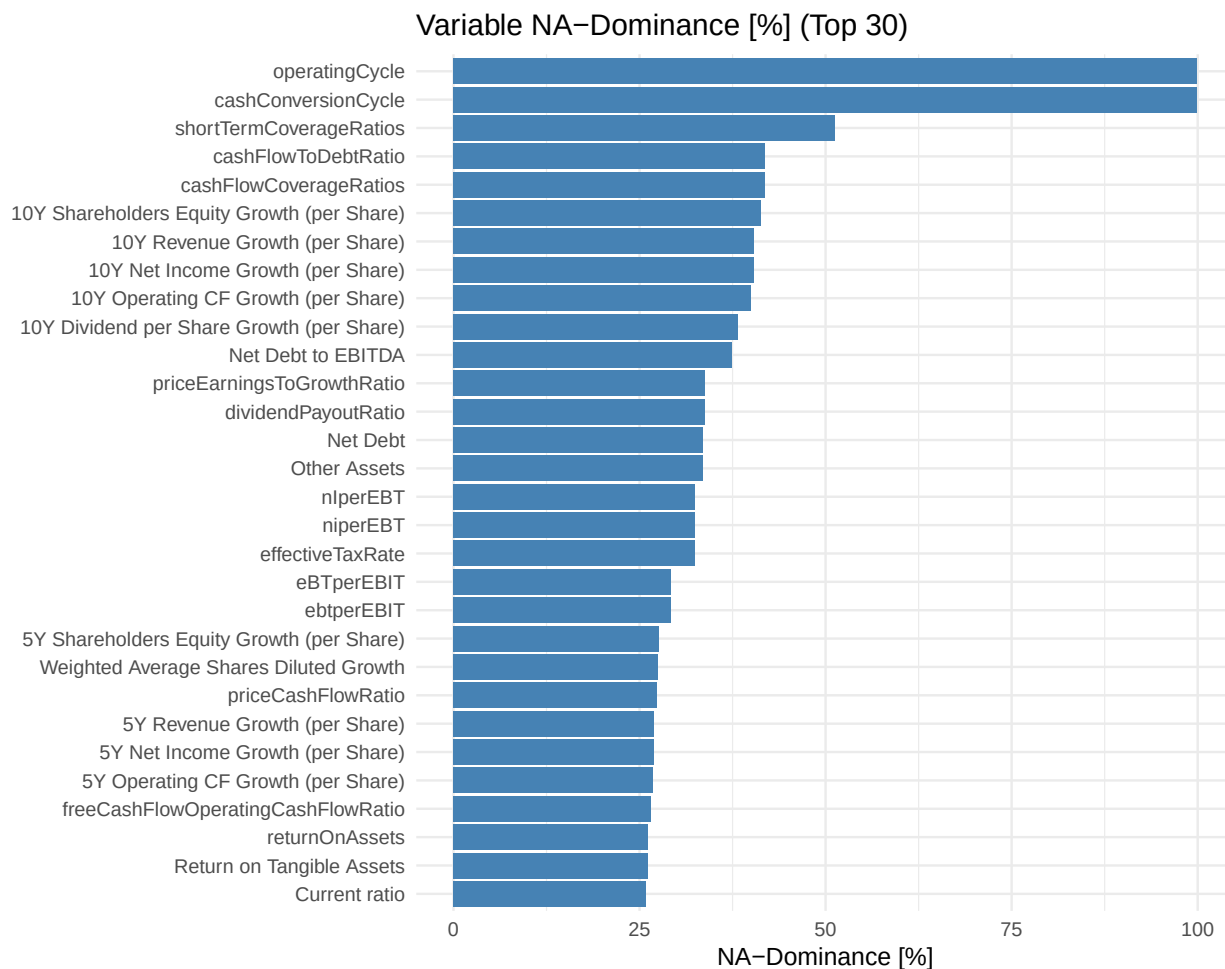
3.1.1 Outlier Removal

A handful of firms show next-year price increases above 500%. These are almost exclusively micro-cap stocks whose returns reflect illiquidity and speculative squeezes rather than fundamentals, and their extreme leverage on squared-error criteria would let a couple dozen firms dominate every variance-based method downstream. We remove them.

```
## Removed 22 firms with >500% next-year price increases
```

3.1.2 Missingness and Zero-Dominance

Many columns are dominated by missing values or exact zeros. Exact zeros in this dataset are frequently a reporting artifact (a metric not applicable to the firm, coded as zero rather than NA), so a zero-dominated column carries little usable variation. We visualize both pathologies and then filter columns using quantile-based thresholds: columns above the 50th percentile of NA-dominance (about 6% missing) and above the 60th percentile of zero-dominance (about 7.7% zeros) are removed.



```
## NA threshold: 6.0% | Zero threshold: 7.7%  
## Columns dropped: 160 | Columns remaining: 65
```

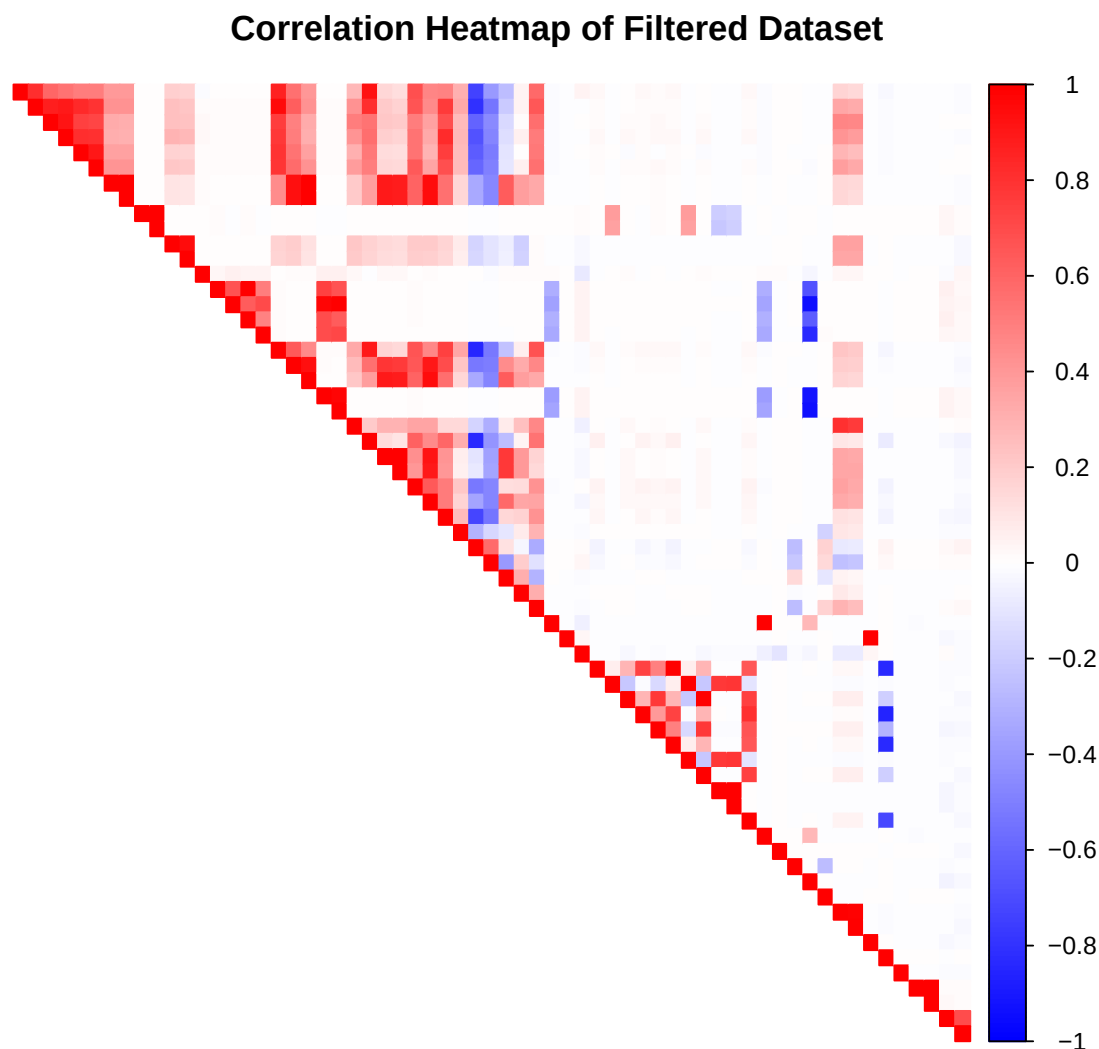
3.1.3 Imputation by Sector and Revenue Similarity

The surviving columns still contain scattered NAs. Dropping incomplete rows would discard firms non-randomly (smaller firms report fewer metrics), so we impute instead. Our imputation respects the structure of the data: a missing value for firm i is replaced by the mean of that metric among firms in the same sector whose revenue lies within 10% of firm i 's revenue. Peer firms of similar scale in the same industry are the natural reference class for a missing accounting figure. Where no revenue-similar peers exist, we fall back to the sector median.

```
## Remaining NAs after imputation: 0
```

3.2 Variable Selection

The filtered dataset still contains dozens of variables that are exact or near-exact linear functions of one another (gross profit and revenue; EBIT and EBITDA; every “per share” variant). The correlation heatmap makes the redundancy visible.



For the factor analysis and inference stages we work with a curated set of key variables chosen to span the main accounting dimensions (scale, profitability, balance sheet position, cash generation, growth, and market valuation) while avoiding mechanical duplicates. Appendix A.1 defines each variable. The PCA and clustering stages use the fuller filtered set, since those methods are meant to discover structure without pre-selection.

```
## [1] 3786 18
```

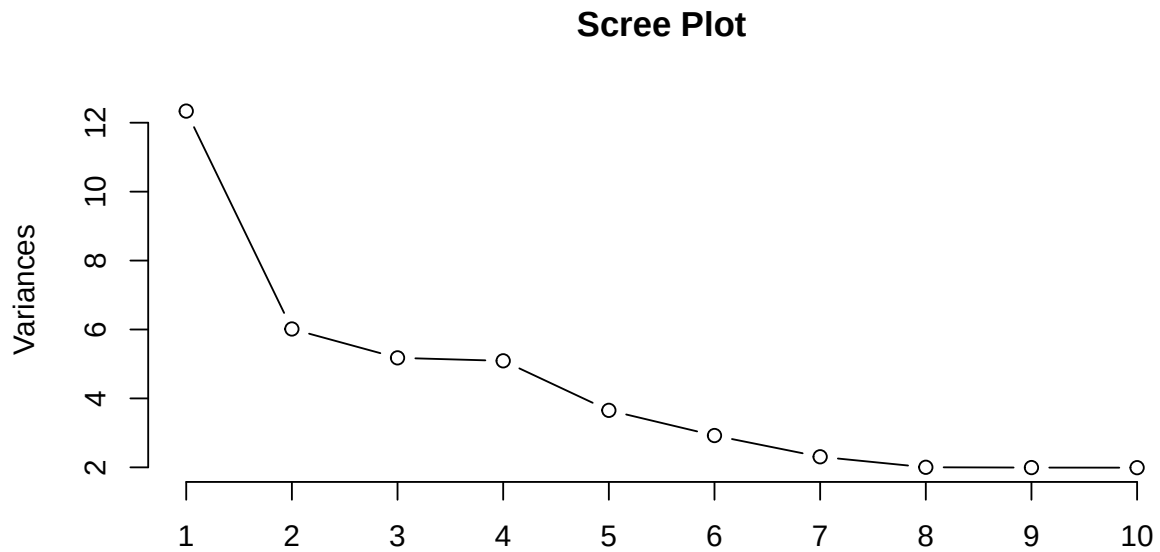
One listed variable, Net Income Growth, does not survive the QA filter, so `any_of()` retains 18 of the 19 names.

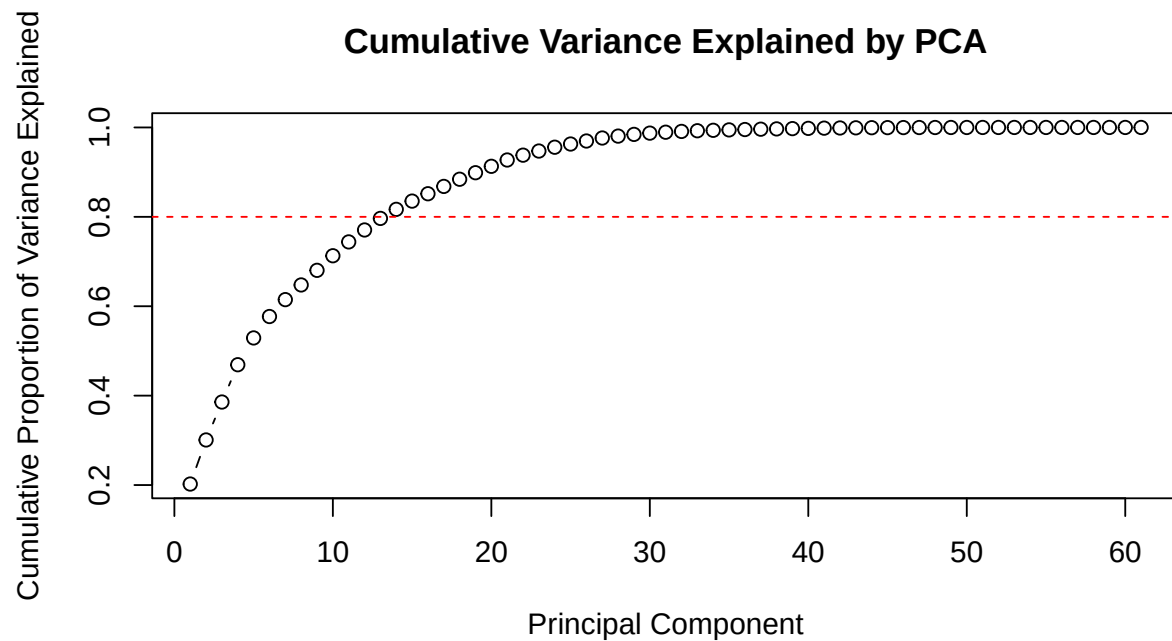
3.3 Principal Component Analysis

We standardize the numeric predictors (excluding the forward return, which is the outcome we want the structure to relate to, and excluding near-zero-variance columns) and run PCA on the correlation matrix.

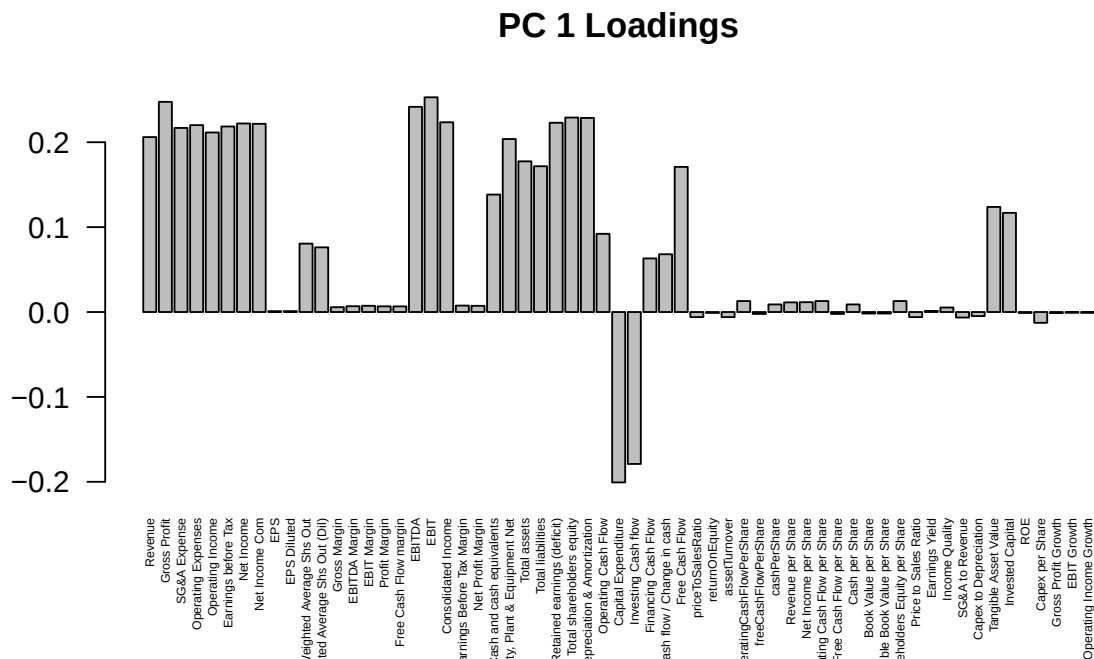
Table 1: Variance explained by the first ten principal components.

PC	Proportion	Cumulative
1	0.202	0.202
2	0.099	0.301
3	0.085	0.386
4	0.083	0.469
5	0.060	0.529
6	0.048	0.577
7	0.038	0.615
8	0.033	0.648
9	0.033	0.680
10	0.033	0.713

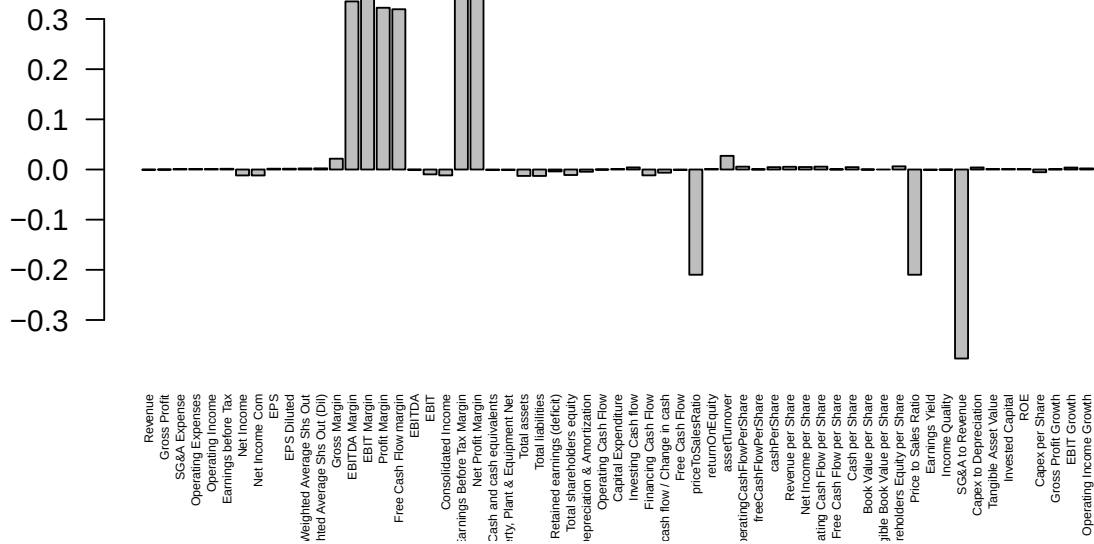




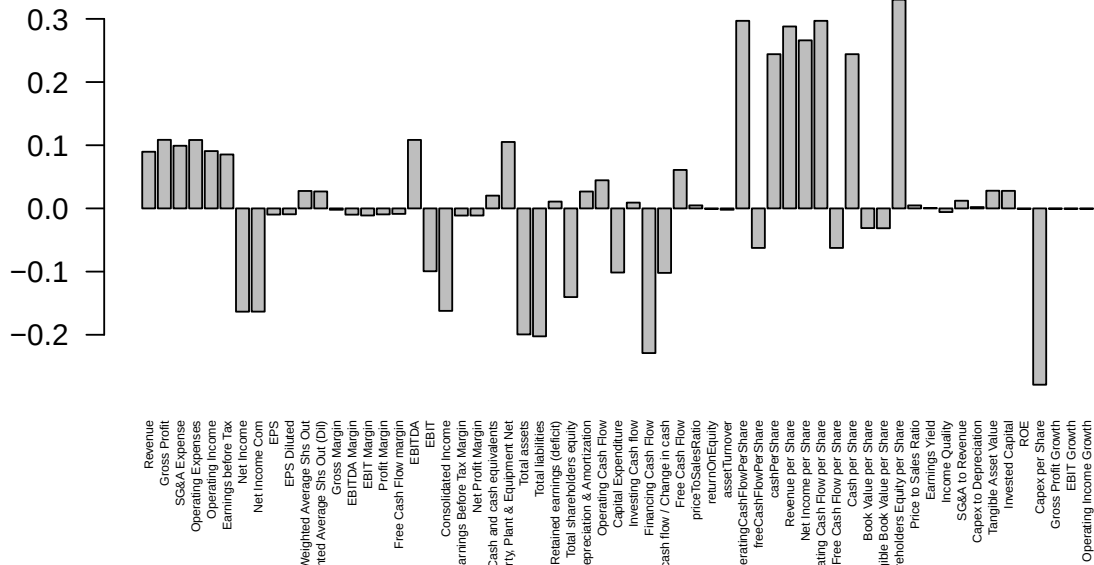
Reaching 80% of the variance would require roughly 16 components, too many to interpret. The scree plot elbows near eight components, which together explain about 65% of the variance, and we retain these eight for the downstream clustering. The gap between 65% and 100% is itself informative: about a third of the standardized variation in these financial metrics is effectively idiosyncratic noise from the perspective of any low-dimensional summary.



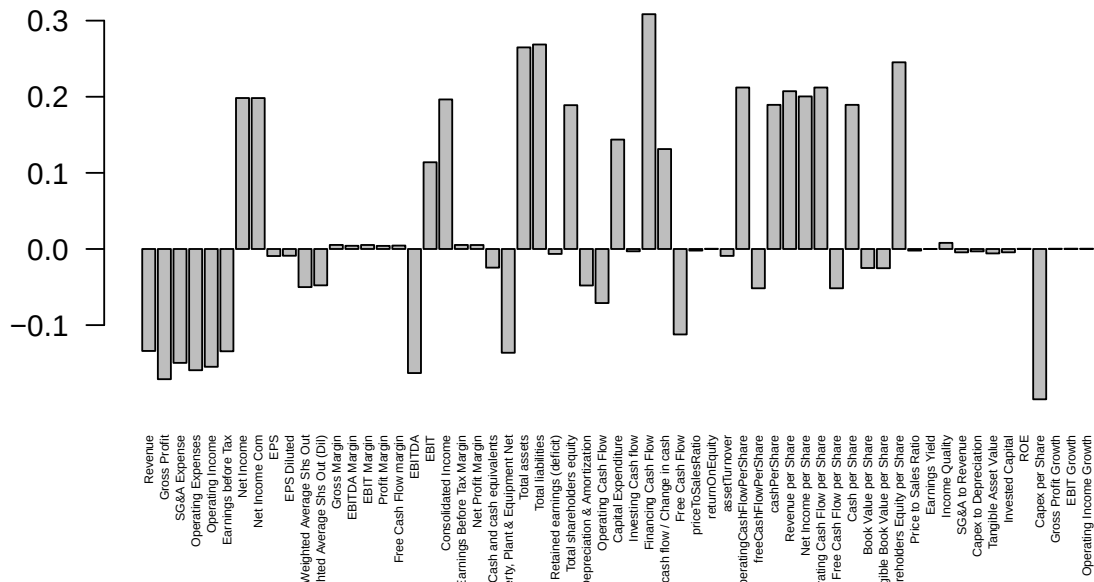
PC 2 Loadings



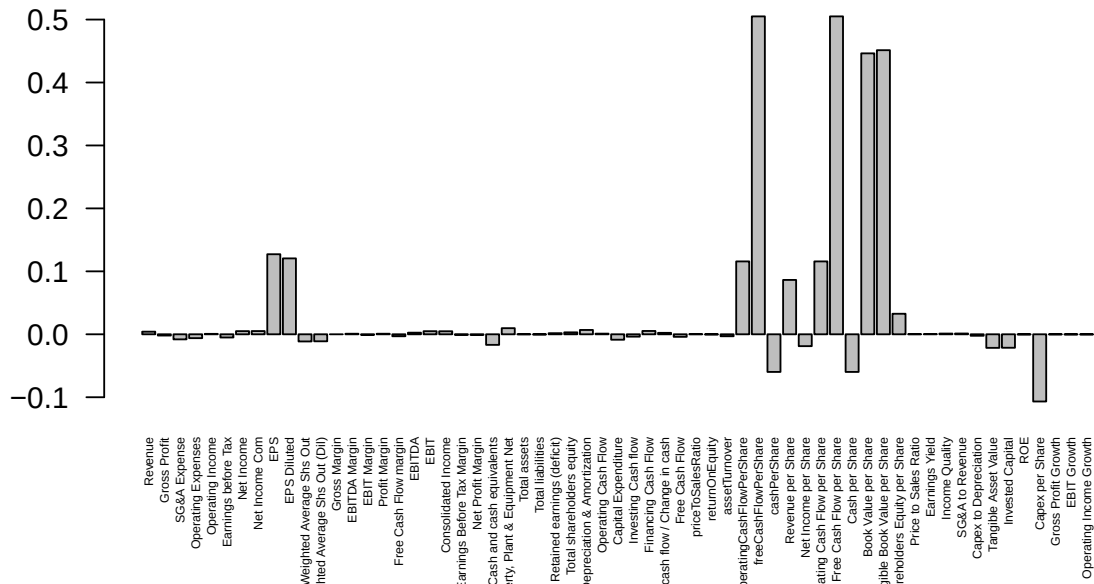
PC 3 Loadings



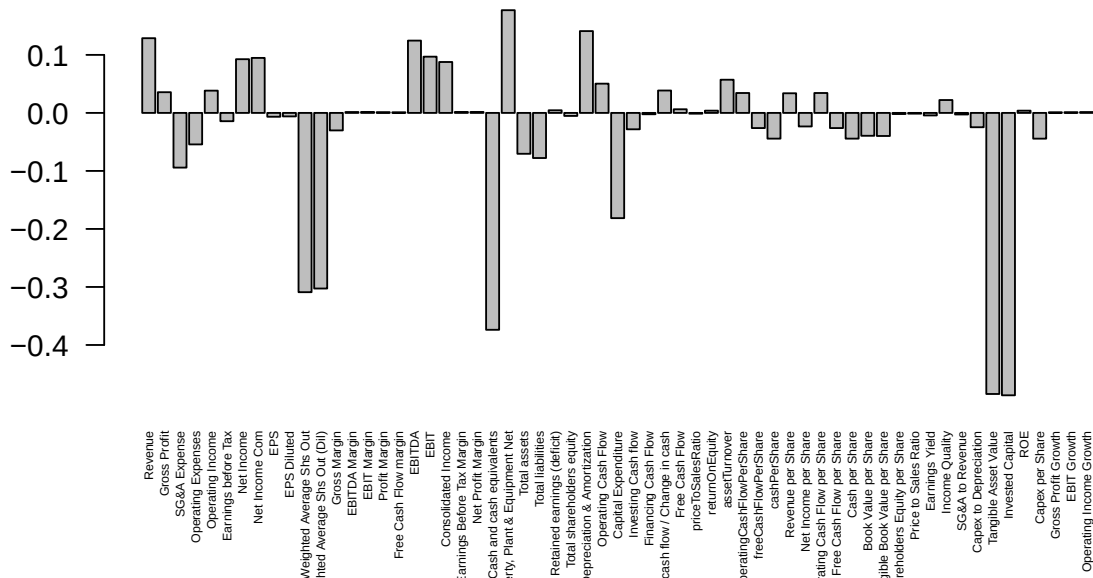
PC 4 Loadings



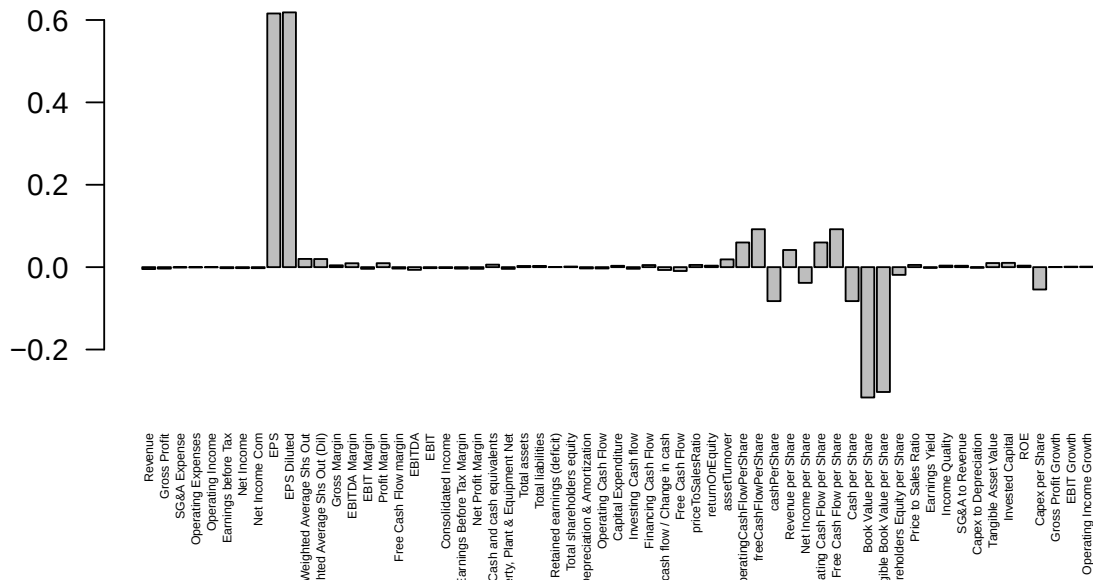
PC 5 Loadings

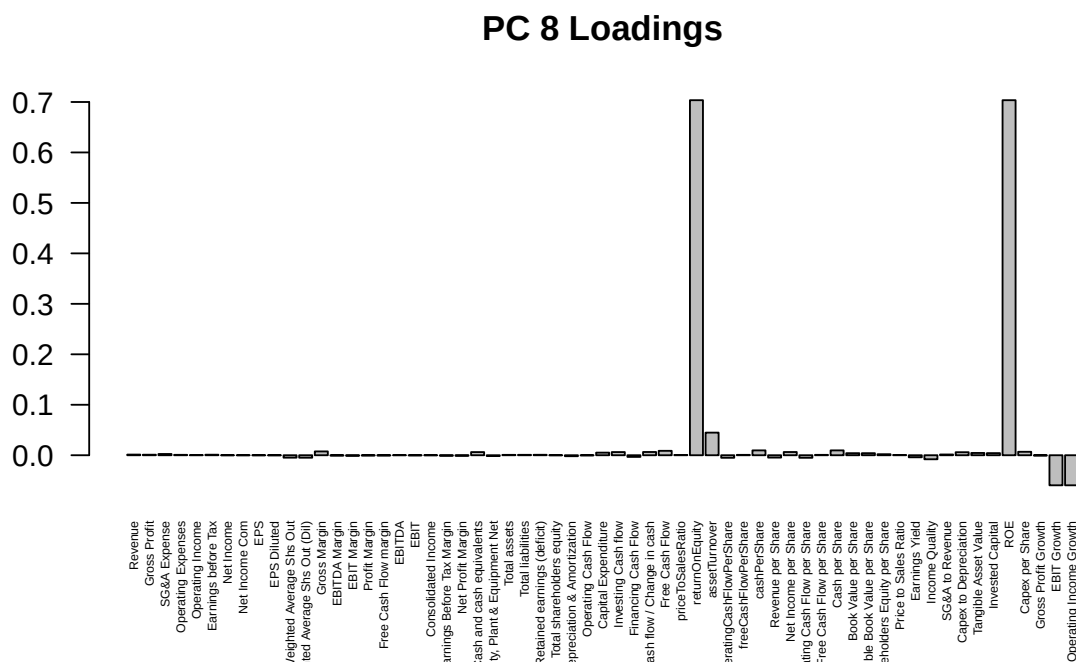


PC 6 Loadings



PC 7 Loadings



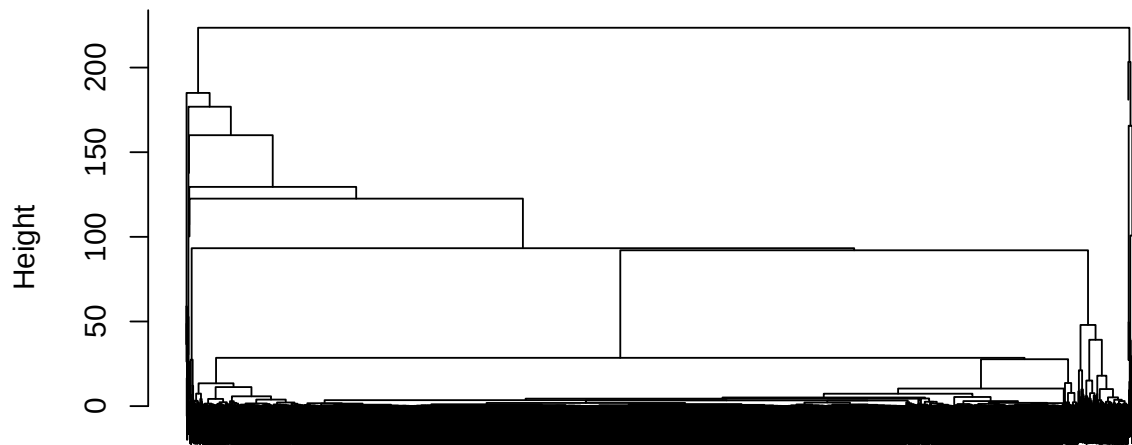


Reading the loadings, PC1 weights income statement and balance sheet levels together against capital expenditure, a general size axis. PC2 loads on margins, PC3 and PC5 on per-share value metrics, and PC4 and PC6 mix cash flow and capitalization measures without a clean label. The final two components expose the dataset's redundancy directly: PC7 is nearly a pure EPS axis and PC8 isolates the two return-on-equity variants, a mechanically duplicated pair the variable filter failed to catch. The axes are suggestive of scale, profitability, and valuation, though several components blend constructs, which motivates the explicit factor model in Section 3.5.

3.4 Clustering Analysis

We cluster firms in the space of the first eight PC scores, comparing Ward's linkage with minimax linkage.

Ward Hierarchical Clustering Dendrogram



##						
##	1	2	3	4	5	6
##	3739	33	8	3	2	1

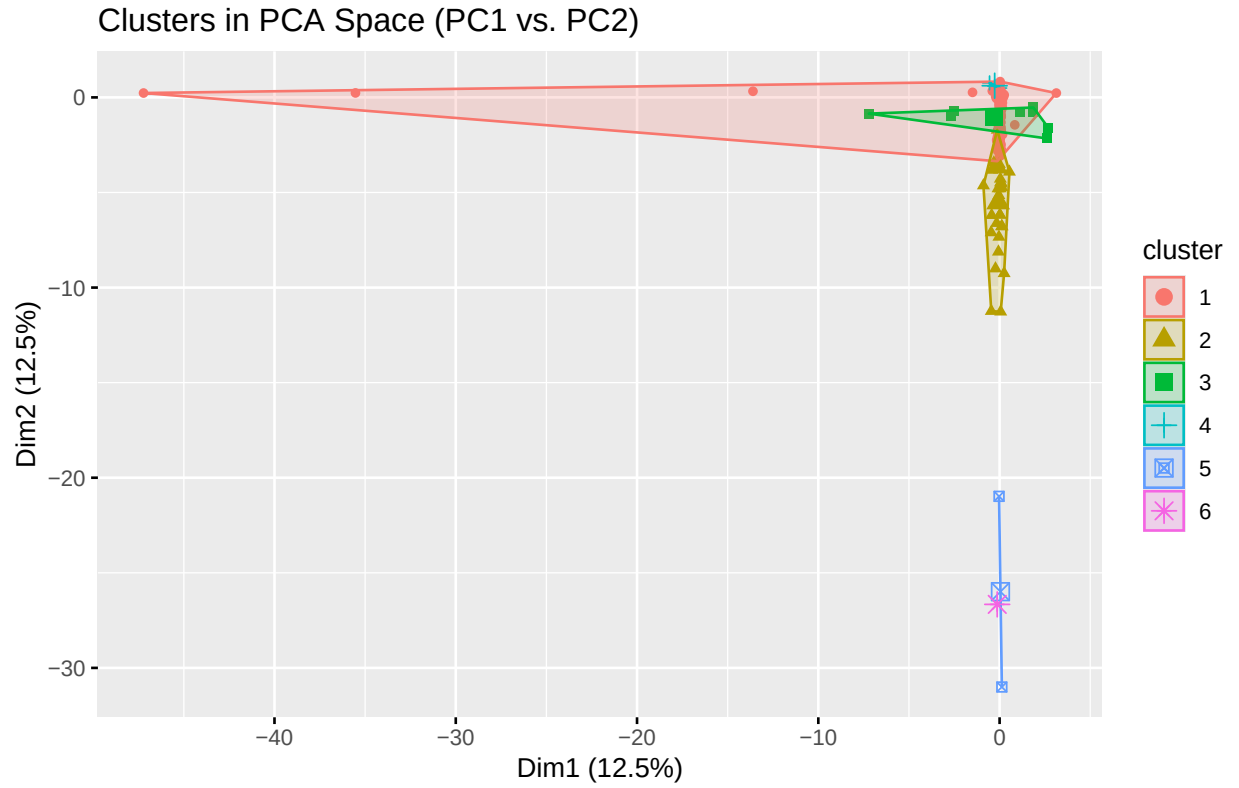
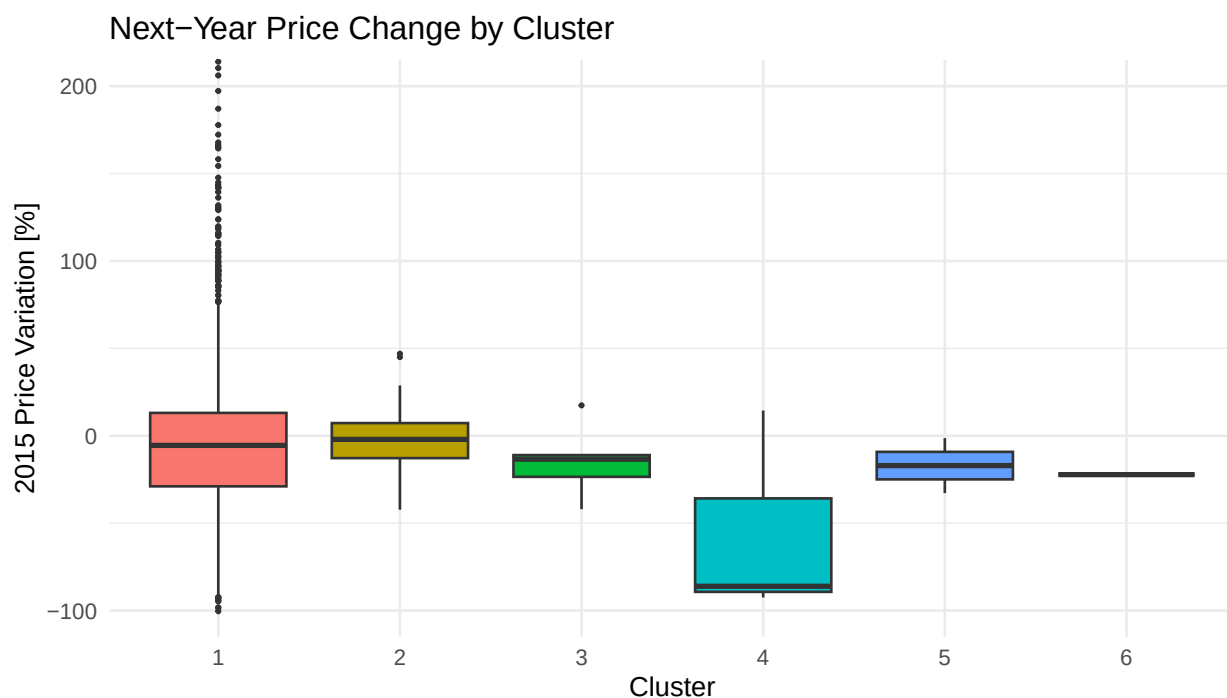
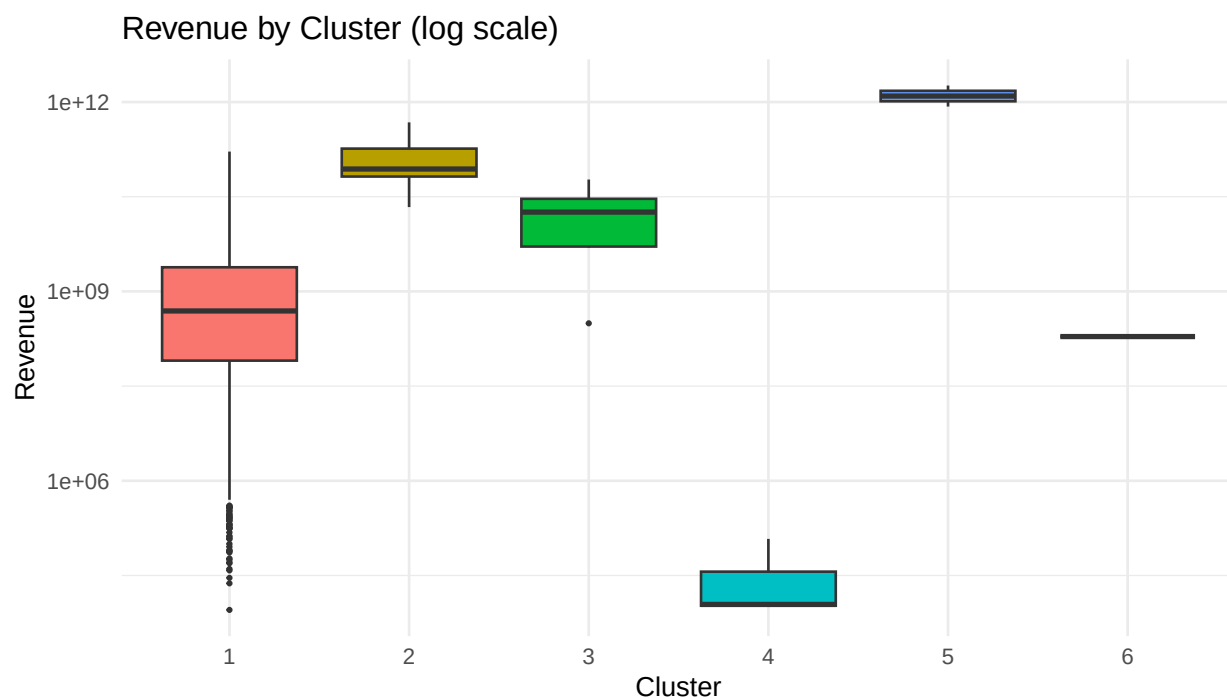


Table 2: Minimax linkage prototype firms for each cluster.

Cluster	Prototype
1	SHG
2	AEZS
3	TOPS
4	AMX
5	MSB
6	ITCB

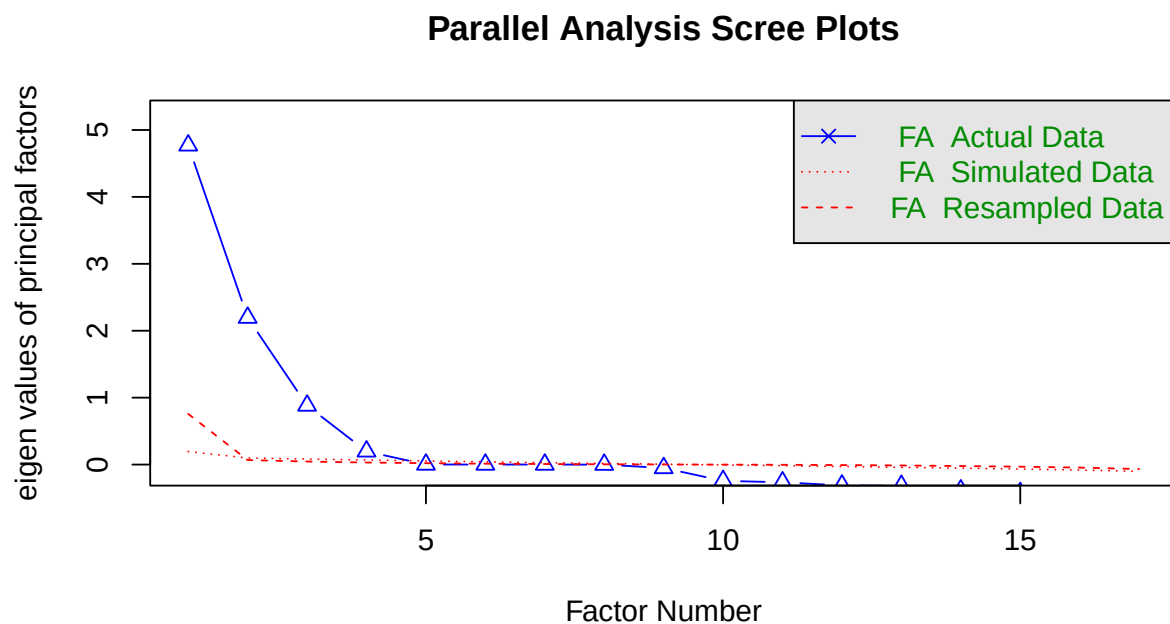


The clustering is degenerate in an informative way: cluster 1 absorbs 3,739 of the 3,786 firms, and the remaining five clusters contain between one and 33. Rather than uncovering firm archetypes, the dendrogram isolates a handful of extreme observations, including firms whose reported revenue exceeds one trillion dollars, values no 2014 filer approached and almost certainly data errors in the source. The tiny clusters thus function as outlier and data-quality detection, and the revenue and price-change boxplots mostly chart the profiles of these anomalous firms rather than differences between comparable groups. This is the heavy-tail pathology

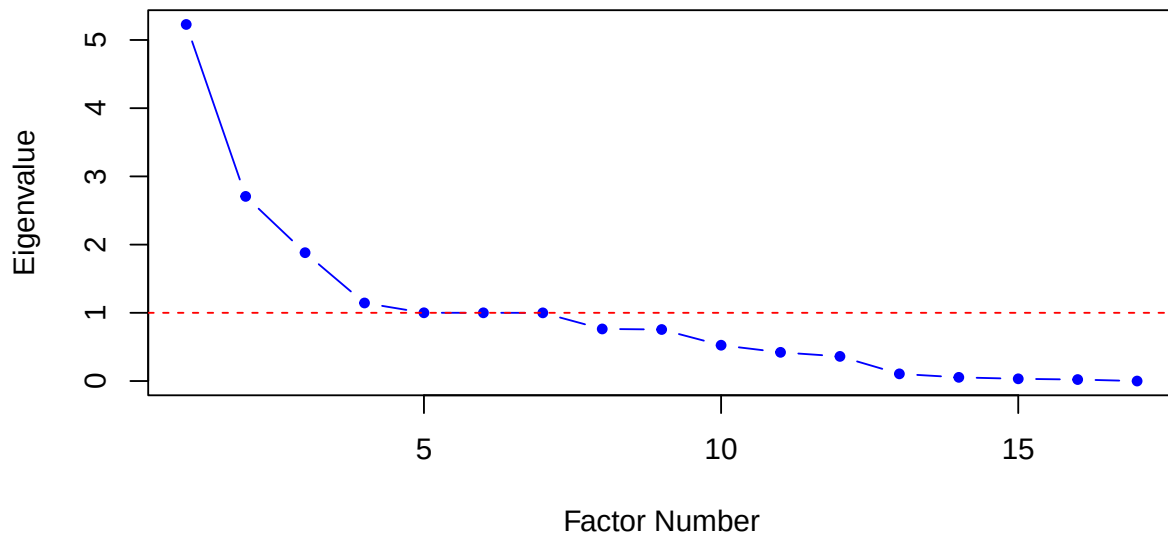
of Section 2.2 made visible: in a space built from variance-maximizing directions, the most extreme firms define the geometry, and even Ward's preference for balanced clusters cannot overcome it. The minimax prototypes tell the same story; the representative firms of the small clusters are thinly traded or anomalously reported names rather than recognizable business archetypes.

3.5 Exploratory Factor Analysis

We now fit the orthogonal factor model on the curated variable set. Parallel analysis and the Kaiser criterion suggest up to six factors, but model complexity and interpretability favor four, though the fourth factor turns out to carry little common variance.

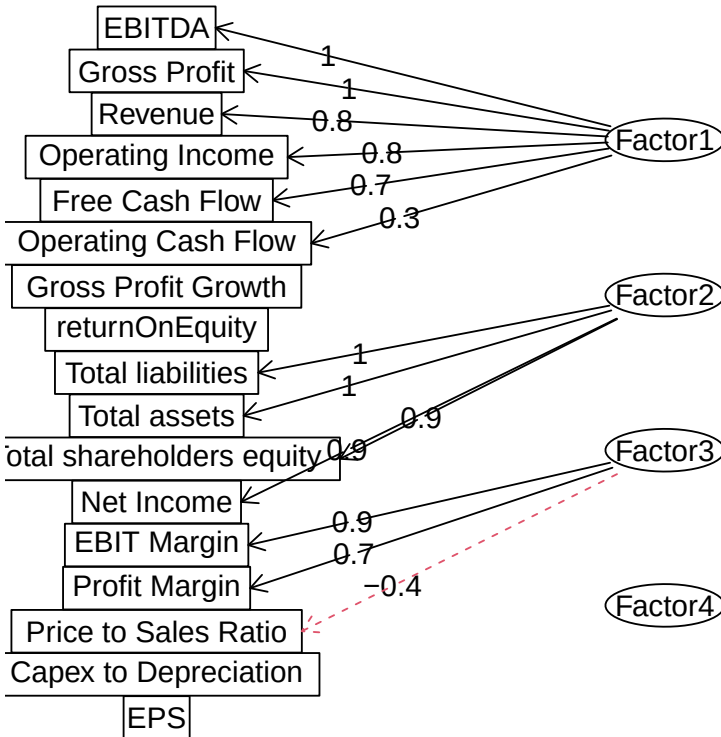


Scree Plot (Key Variables)



```
##
## Loadings:
##
## Factor1 Factor2 Factor3 Factor4
## Revenue      0.848              0.419
## Gross Profit  0.957
## Operating Income 0.820          -0.479
## Net Income    0.351  0.879
## EPS
## EBITDA        0.978
## EBIT Margin           0.870
## Profit Margin      0.725
## Free Cash Flow    0.663
## returnOnEquity
## Price to Sales Ratio           -0.412
## Total assets           0.995
## Total liabilities      0.996
## Total shareholders equity 0.366  0.895
## Operating Cash Flow    0.345
## Capex to Depreciation
## Gross Profit Growth
##
## Factor1 Factor2 Factor3 Factor4
## SS loadings  4.092  3.619  1.487  0.439
## Proportion Var 0.241  0.213  0.087  0.026
## Cumulative Var 0.241  0.454  0.541  0.567
```

EFA Factor Structure



The varimax-rotated loadings support three strong latent constructs and one weak residual:

- **Factor 1, operational scale.** EBITDA (0.98), gross profit (0.96), revenue (0.85), operating income (0.83), free cash flow (0.66). Income-statement magnitude: how big the firm's operations are.
- **Factor 2, balance sheet position.** Total assets (1.00), total liabilities (1.00), shareholders' equity (0.89), and net income (0.88). The balance sheet moves as a bloc, and notably the bottom line sits here rather than with the operating-scale factor.
- **Factor 3, profitability.** EBIT margin (0.87), profit margin (0.73), with a negative loading on price-to-sales (-0.41): more profitable firms in this sample trade at lower sales multiples.
- **Factor 4, a weak residual.** A contrast between revenue (0.43) and operating income (-0.47) carrying only 2.4% of the variance, suggestive of cost-structure differences but too faint to interpret with confidence.

Two features of the solution deserve comment. First, the separation of scale (Factor 1) from the balance sheet (Factor 2) is substantively interesting: income statement magnitude and balance sheet magnitude are correlated but load on distinct factors, suggesting that leverage and capital structure vary meaningfully across firms of similar operating size. Second, four variables (EPS, return on equity, capex-to-depreciation, and gross profit growth) have uniquenesses of essentially 1.00: after standardization they correlate with nothing else, a signature of the extreme outliers that dominate their scaled variance and a preview of the heavy-tail mechanism Simulation 2 isolates.

3.6 Confirmatory Factor Analysis

To test whether the two dominant constructs survive as an explicit hypothesis, we specify a two-factor CFA with operational scale measured by revenue, gross profit, and operating income, and financial stability

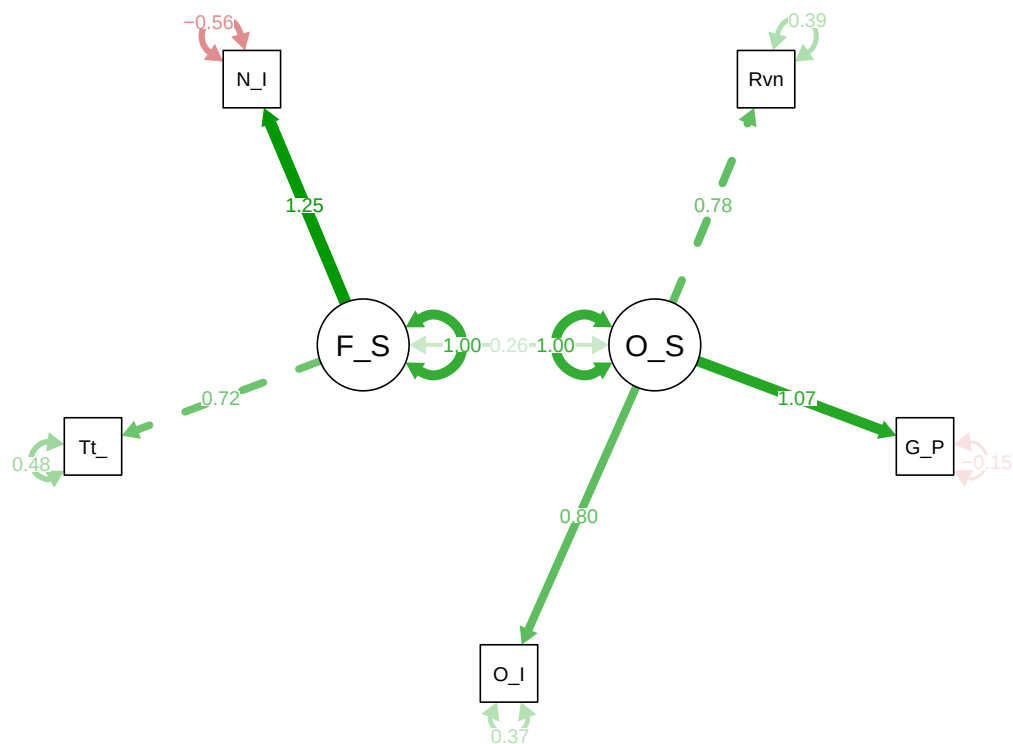
measured by total assets and net income.

```
## lavaan 0.6-19 ended normally after 36 iterations
##
##      Estimator                      ML
##      Optimization method          NLMINB
##      Number of model parameters      11
##
##      Number of observations          3786
##
## Model Test User Model:
##
##      Test statistic                  2270.188
##      Degrees of freedom                4
##      P-value (Chi-square)            0.000
##
## Model Test Baseline Model:
##
##      Test statistic                  19159.297
##      Degrees of freedom                10
##      P-value                          0.000
##
## User Model versus Baseline Model:
##
##      Comparative Fit Index (CFI)      0.882
##      Tucker-Lewis Index (TLI)        0.704
##
## Loglikelihood and Information Criteria:
##
##      Loglikelihood user model (H0)    -18413.452
##      Loglikelihood unrestricted model (H1) -17278.358
##
##      Akaike (AIC)                    36848.903
##      Bayesian (BIC)                   36917.533
##      Sample-size adjusted Bayesian (SABIC) 36882.580
##
## Root Mean Square Error of Approximation:
##
##      RMSEA                          0.387
##      90 Percent confidence interval - lower 0.374
##      90 Percent confidence interval - upper 0.400
##      P-value H_0: RMSEA <= 0.050        0.000
##      P-value H_0: RMSEA >= 0.080        1.000
##
## Standardized Root Mean Square Residual:
##
##      SRMR                          0.057
##
## Parameter Estimates:
##
##      Standard errors                  Standard
##      Information                      Expected
##      Information saturated (h1) model  Structured
##
```

```

## Latent Variables:
##
##           Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## Operational_Scale =~
##   Revenue           1.000
##   Gross_Profit       1.375    0.019  71.959   0.000   1.070   1.070
##   Operating_Incm     1.022    0.017  59.484   0.000   0.796   0.796
## Financial_Stability =~
##   Total_assets       1.000
##   Net_Income         1.737    0.072  24.252   0.000   1.249   1.249
##
## Covariances:
##           Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## Operational_Scale ~~
##   Financl_Stblty     0.148    0.012  12.508   0.000   0.264   0.264
##
## Variances:
##           Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## .Revenue         0.394    0.010  41.253   0.000   0.394   0.394
## .Gross_Profit     -0.146    0.008 -18.398   0.000  -0.146  -0.146
## .Operating_Incm   0.367    0.009  40.417   0.000   0.367   0.367
## .Total_assets     0.483    0.023  21.308   0.000   0.483   0.483
## .Net_Income       -0.560    0.061  -9.181   0.000  -0.560  -0.560
## Operationl_Scl    0.606    0.021  28.384   0.000   1.000   1.000
## Financl_Stblty    0.517    0.028  18.335   0.000   1.000   1.000

```



The fitted model produces $CFI \approx 0.88$, $RMSEA \approx 0.39$, and $SRMR \approx 0.06$. The SRMR is acceptable, but the CFI falls short of the 0.95 benchmark and the RMSEA is far above any conventional threshold. The loadings themselves are strong (all standardized loadings above 0.7), and some exceed 1 in unstandardized form alongside negative residual variances, a Heywood case signaling that the model is misspecified or that the latent factors are nearly collinear. The confirmatory verdict is clear: while the exploratory factors label the data helpfully, the strict two-factor structure does not reproduce the observed covariance. Financial metrics carry too many cross-loadings and too much mechanical dependence for a simple-structure model to fit.

3.7 Sector-Level Inference

3.7.1 MANOVA

Do financial profiles differ across sectors when tested jointly? We fit a one-way MANOVA with 13 key metrics as the multivariate response and sector as the grouping factor.

```
##           Df    Wilks approx F num Df den Df    Pr(>F)
## Sector      10 0.86534   4.2269     130 30120 < 2.2e-16 ***
## Residuals 3775
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Response Revenue :
##           Df    Sum Sq    Mean Sq F value    Pr(>F)
## Sector      10 8.3441e+22 8.3441e+21  5.5792 2.611e-08 ***
## Residuals 3775 5.6459e+24 1.4956e+21
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response Net_Income :
##           Df    Sum Sq    Mean Sq F value    Pr(>F)
## Sector      10 4.7188e+20 4.7188e+19  2.4462 0.006586 **
## Residuals 3775 7.2821e+22 1.9290e+19
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response Total_assets :
##           Df    Sum Sq    Mean Sq F value    Pr(>F)
## Sector      10 6.2974e+24 6.2974e+23  5.0008 2.99e-07 ***
## Residuals 3775 4.7538e+26 1.2593e+23
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response Gross_Profit :
##           Df    Sum Sq    Mean Sq F value    Pr(>F)
## Sector      10 1.7529e+22 1.7529e+21 13.588 < 2.2e-16 ***
## Residuals 3775 4.8699e+23 1.2900e+20
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response EBITDA :
##           Df    Sum Sq    Mean Sq F value    Pr(>F)
## Sector      10 3.9285e+21 3.9285e+20 11.115 < 2.2e-16 ***
```

```

## Residuals    3775 1.3343e+23 3.5345e+19
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response Gross_Profit_Growth :
##           Df      Sum Sq Mean Sq F value Pr(>F)
## Sector      10 2.2285e+08 22284616  0.6997 0.7257
## Residuals   3775 1.2023e+11 31849522
##
## Response Operating_Cash_Flow :
##           Df      Sum Sq Mean Sq F value Pr(>F)
## Sector      10 7.0880e+21 7.088e+20  2.5233 0.005022 **
## Residuals   3775 1.0604e+24 2.809e+20
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response EPS :
##           Df      Sum Sq Mean Sq F value Pr(>F)
## Sector      10 1.5831e+13 1.5831e+12  0.5728 0.8374
## Residuals   3775 1.0434e+16 2.7639e+12
##
## Response EBIT_Margin :
##           Df      Sum Sq Mean Sq F value Pr(>F)
## Sector      10  171343 17134.3  4.3579 4.26e-06 ***
## Residuals   3775 14842333  3931.7
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response Profit_Margin :
##           Df      Sum Sq Mean Sq F value Pr(>F)
## Sector      10  457129  45713  5.0193 2.767e-07 ***
## Residuals   3775 34380523  9107
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Response returnOnEquity :
##           Df      Sum Sq Mean Sq F value Pr(>F)
## Sector      10 6.2085e+10 6.2085e+09  0.4765 0.9062
## Residuals   3775 4.9188e+13 1.3030e+10
##
## Response Capex_to_Depreciation :
##           Df      Sum Sq Mean Sq F value Pr(>F)
## Sector      10  5387127  538713  1.2902 0.2297
## Residuals   3775 1576244228  417548
##
## Response Price_to_Sales_Ratio :
##           Df      Sum Sq Mean Sq F value Pr(>F)
## Sector      10 19221859 1922186  5.0302 2.644e-07 ***
## Residuals   3775 1442537849  382129
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Wilks' lambda rejects equality of sector mean vectors decisively. The follow-up univariate ANOVAs show significant sector differences for most metrics, with four exceptions: gross profit growth, EPS, return on

equity, and capex-to-depreciation. The pattern is coherent. Level variables (revenue, assets, costs, total profit) differ strongly by industry, while several normalized efficiency and growth measures do not, suggesting that sectors differ more in scale and structure than in per-dollar performance.

3.7.2 Hotelling’s T^2 : Up Firms vs. Down Firms

Finally, we ask whether firms whose stock rose the following year differ from firms whose stock fell, on the full vector of metrics simultaneously.

```
## Test stat: 12.303
## Numerator df: 13
## Denominator df: 3772
## P-value: 0.5063
```

The test comes back null ($p \approx 0.51$): on this 13-metric profile, firms whose stock subsequently rose are statistically indistinguishable from firms whose stock fell. The contrast with the MANOVA is instructive. The same machinery detects sector separation decisively on the same metrics, so the null here does not reflect a lack of power; it reflects the well-known difficulty of predicting returns from public accounting data. Whatever information 10-K fundamentals carry about next-year price movement, it does not appear as a mean shift in the joint financial profile.

3.8 Limitation Analysis

Four limitations qualify everything above.

Heavy tails and outlier leverage. Financial ratios have extremely heavy-tailed distributions, and every method here is variance-based. We removed the most egregious price outliers, but the remaining tails still give a small number of firms disproportionate influence on the leading eigenvectors, the cluster geometry, and the MANOVA covariance estimates. Simulation 2 in the next section quantifies this mechanism directly.

Mechanical collinearity. Many metrics are deterministic functions of others. This inflates the apparent strength of the leading factors (Factor 2’s loadings of 1.00 on assets and liabilities reflect the accounting identity, not a discovered construct) and it produces the near-singular covariance that generates Heywood cases in the CFA.

Assumption violations in inference. MANOVA and Hotelling’s T^2 assume homogeneous within-group covariance. Sectors as different as utilities and healthcare almost certainly violate this, and while large samples protect the type I error rate to a degree, the tests should be read as descriptive evidence of separation rather than exact inference.

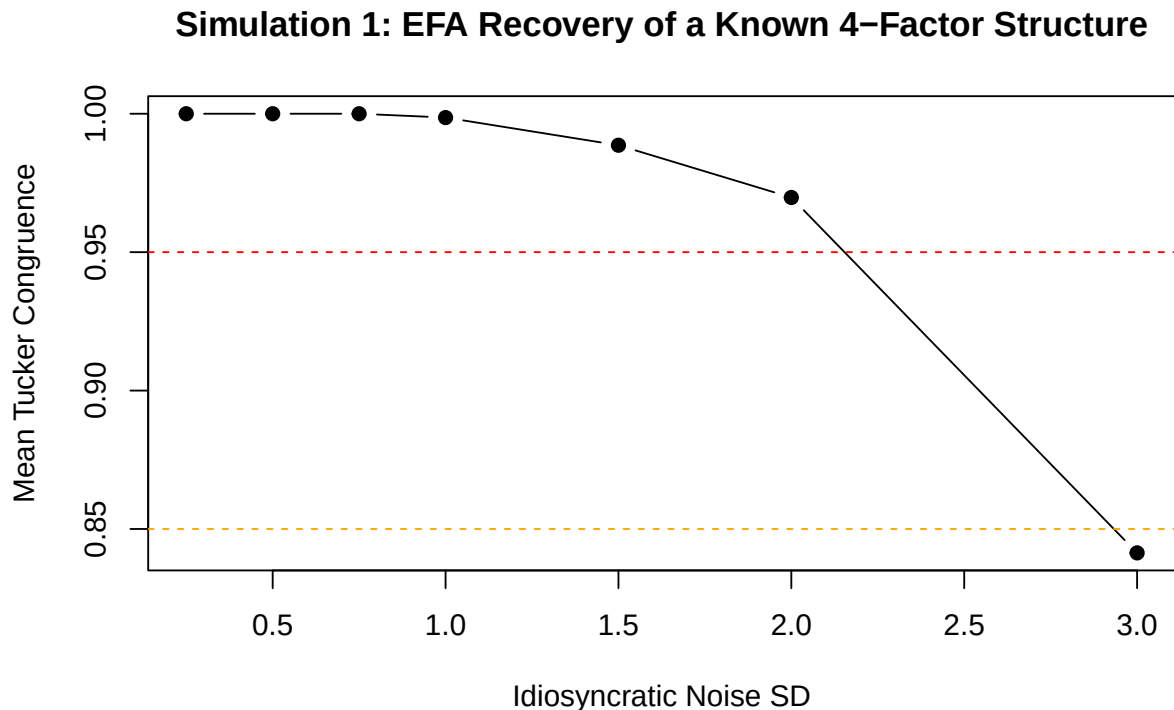
Imputation and selection effects. The quantile-based column filtering and the sector-revenue imputation both encode choices. Filtering removes metrics that are informative precisely for the firms that report them (e.g., R&D-heavy firms), and imputation shrinks missing firms toward their sector peers, which mildly inflates the sector differences MANOVA later detects. The direction of this bias is worth remembering when interpreting the sector results.

4 Simulations

The 10-K data gave us a useful setting for thinking through the uses and challenges of latent-variable methods, though the preceding sections expose the limits of applying them to heavy-tailed, collinear accounting data. In the data analysis we could never observe the true latent structure, so we could not distinguish “the model found the wrong factors” from “the data contain no clean factors.” Simulation removes this ambiguity. We generate data from a known factor model, so recovery can be measured directly, and we then degrade the data in the two ways the limitation analysis identified: idiosyncratic noise and heavy tails.

4.1 Recovering a Known Factor Structure

The first simulation asks how well maximum likelihood EFA recovers a known loading matrix as idiosyncratic noise grows. We build a four-factor, 16-variable population loading matrix with clean simple structure (mimicking the scale / stability / profitability / growth pattern from the data analysis), generate observations from $\mathbf{x} = \mathbf{\Lambda}\mathbf{f} + \sigma\boldsymbol{\varepsilon}$, and measure recovery by Tucker's congruence coefficient between the estimated and true loadings, matched factor by factor. Congruence above 0.95 is conventionally read as factor equality; below 0.85, as failure to recover.

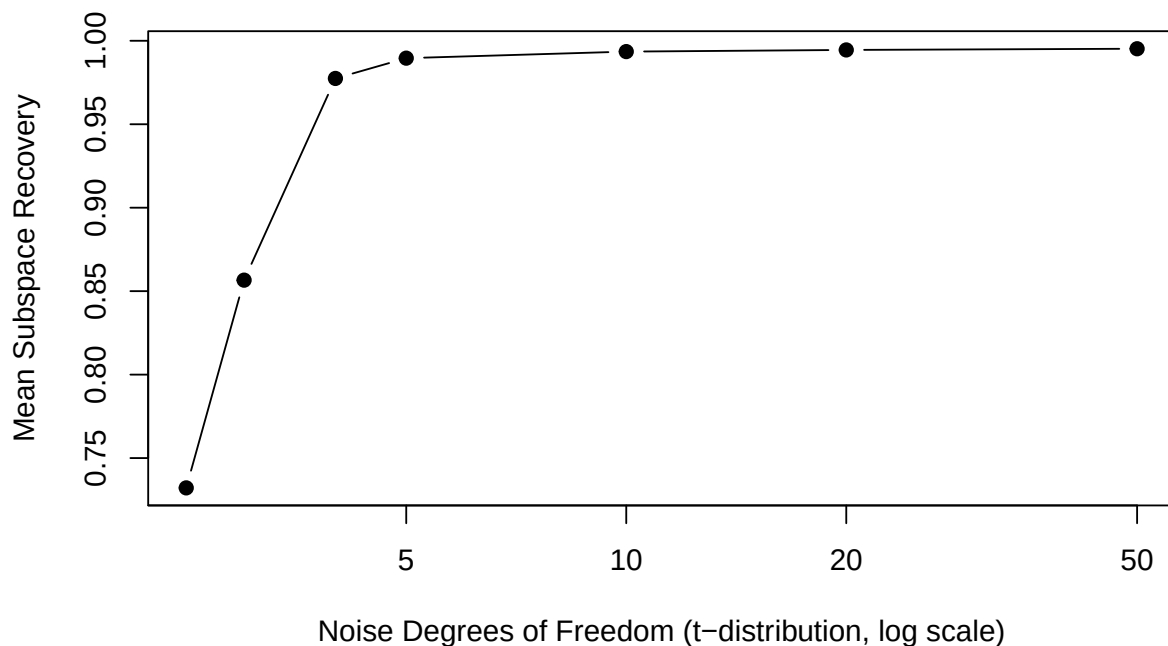


With modest noise the four factors are recovered nearly perfectly (congruence above 0.98), and recovery degrades gracefully as uniquenesses grow. The pattern clarifies the data-analysis results: when a clean factor structure exists, EFA finds it, so the interpretability problems we encountered on the 10-K data reflect the data (cross-loadings, mechanical identities, heavy tails) rather than a failure of the method.

4.2 PCA Under Heavy Tails

The second simulation targets the pathology most specific to financial data. We generate observations with a true two-dimensional principal subspace plus noise drawn from a t_ν distribution, and vary the degrees of freedom ν from effectively Gaussian ($\nu = 50$) down to the very heavy tails typical of financial ratios ($\nu = 2.5$). Recovery is measured by the average squared canonical correlation between the true and estimated two-dimensional subspaces (1 indicates perfect recovery).

Simulation 2: PCA Subspace Recovery Under Heavy Tails



With Gaussian-like noise the true subspace is recovered almost exactly. As the tails thicken, recovery deteriorates, and below $\nu \approx 3$, where fourth moments diverge, the leading eigenvectors are increasingly captured by individual extreme observations rather than the true structure. This is the mechanism behind the 65% variance ceiling and the unstable clusters in Section 3: a meaningful share of the leading “variance” in financial data belongs to tail events, and PCA cannot tell the difference. The simulation also justifies our outlier filtering, and points toward robust PCA variants as a natural refinement.

Together the simulations show what these methods can do when their core assumptions hold, and they localize the difficulties in the data analysis to specific, measurable violations rather than to the methods themselves.

5 Conclusion

Multivariate analysis of 10-K financial data recovers a coherent low-dimensional story: firms vary along interpretable axes of operational scale, balance sheet position, and profitability, and sectors differ significantly in their joint financial profiles, even as rising and falling stocks remain jointly indistinguishable on the same metrics. At the same time, every stage of the analysis ran into the same wall. Roughly a third of the variance resists any low-dimensional summary, the confirmatory model fits poorly, and the clustering collapses into one mass cluster plus a handful of extreme firms. Our simulations locate the cause in the data’s heavy tails, mechanical collinearity, and idiosyncratic noise rather than in the methods.

Several directions follow naturally. Restricting to a single sector would remove the dominant source of covariance heterogeneity and let the factor model target within-industry structure. Robust or sparse PCA variants would address the tail sensitivity that Simulation 2 demonstrates. And the panel dimension of the data (five filing years per firm) invites longitudinal treatment: do the latent factors persist within firms over time, and does movement along the profitability factor predict subsequent returns better than any single ratio? The question this project leaves open is whether the latent structure in financial statements is stable

enough to trade on, or whether it is, like so much else in this dataset, mostly noise wearing the costume of structure.

6 Works Cited

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Appendix

A.1 Variable Definitions

Financial Term	Brief Description
Revenue	Total money earned from primary business activities before expenses
Gross Profit	Revenue minus cost of goods sold
Operating Income	Profit from core operations; gross profit minus operating expenses
Net Income	Bottom-line profit after all expenses, taxes, and costs
EPS	Profit allocated to each outstanding common share
EBITDA	Earnings before interest, taxes, depreciation, and amortization
EBIT Margin	EBIT divided by revenue
Profit Margin	Net income divided by revenue
Free Cash Flow	Cash generated after operating and capital expenditures
returnOnEquity	Net income relative to shareholders' equity
Price to Sales Ratio	Stock price relative to revenue per share
Total assets	Total value of all resources owned by the firm
Total liabilities	All financial obligations and debts owed
Total shareholders equity	Assets minus liabilities
Operating Cash Flow	Cash generated from normal operations
Capex to Depreciation	Capital expenditures relative to depreciation
Gross Profit Growth	Period-over-period growth in gross profit
Net Income Growth	Period-over-period growth in net income (removed by the QA filter)
2015 PRICE VAR [%]	Percent change in stock price over 2015 (forward outcome)

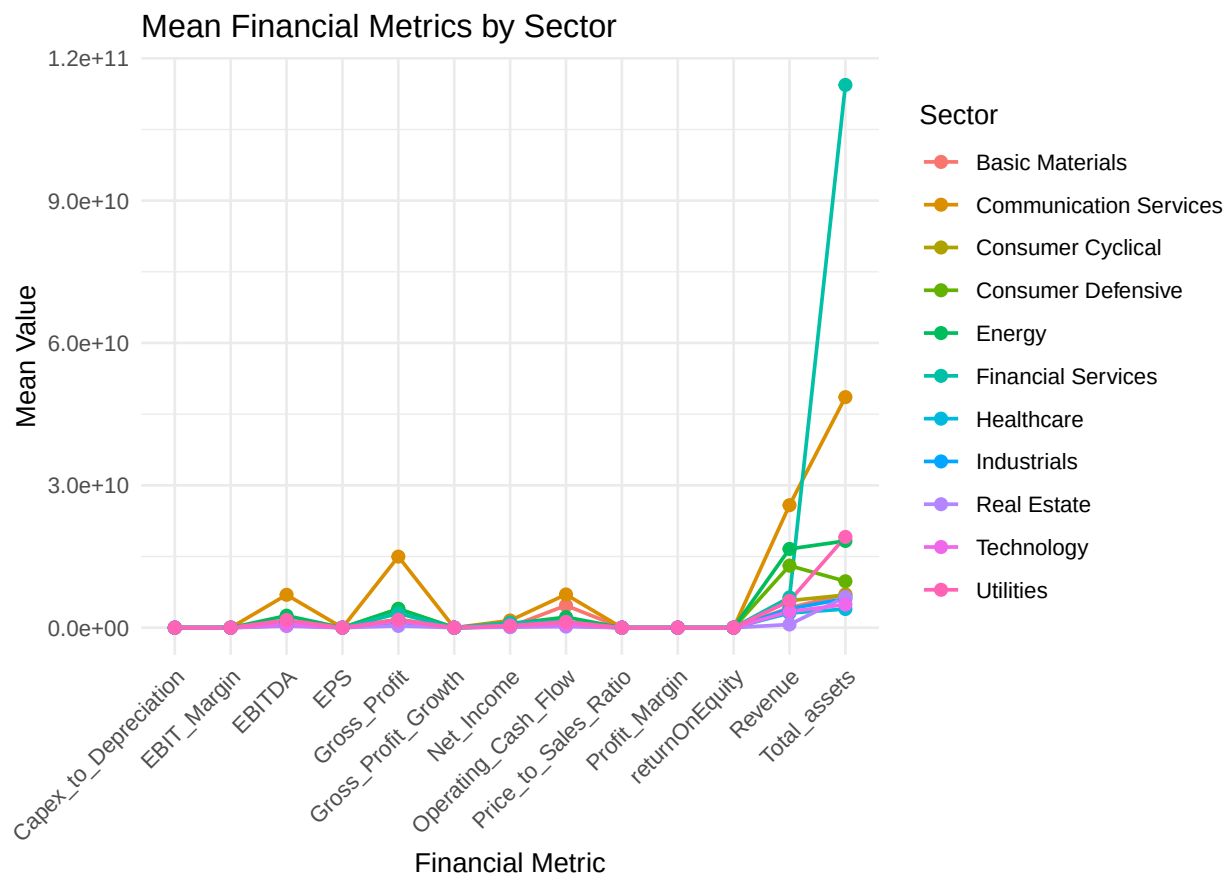
A.2 Full EFA Loadings and Uniquenesses

```
##
## Loadings:
##          Factor1 Factor2 Factor3 Factor4
## Revenue      0.848   0.121   0.071   0.419
## Gross Profit  0.957   0.135   0.086  -0.071
## Operating Income 0.820   0.060   0.081  -0.479
## Net Income    0.351   0.879   0.035   0.007
## EPS           0.003   0.001   0.003   0.000
## EBITDA        0.978   0.106   0.086   0.045
## EBIT Margin   -0.066   0.001   0.870   0.009
## Profit Margin -0.053   0.001   0.725   0.007
## Free Cash Flow 0.663   0.121   0.058   0.032
## returnOnEquity -0.003  -0.001   0.001   0.000
## Price to Sales Ratio 0.025  -0.002  -0.412  -0.003
## Total assets   0.065   0.995   0.010  -0.048
## Total liabilities 0.039   0.996   0.008  -0.065
## Total shareholders equity 0.366   0.895   0.036   0.137
## Operating Cash Flow 0.345   0.043   0.031   0.025
## Capex to Depreciation 0.001   0.000   0.010   0.000
## Gross Profit Growth -0.004  -0.001   0.001   0.001
##
##          Factor1 Factor2 Factor3 Factor4
## SS loadings    4.092   3.619   1.487   0.439
```

```
## Proportion Var    0.241    0.213    0.087    0.026
## Cumulative Var    0.241    0.454    0.541    0.567
```

```
##              Revenue              Gross Profit              Operating Income
##              0.085              0.054              0.089
##              Net Income              EPS              EBITDA
##              0.103              1.000              0.023
##              EBIT Margin              Profit Margin              Free Cash Flow
##              0.238              0.471              0.542
##              returnOnEquity              Price to Sales Ratio              Total assets
##              1.000              0.830              0.005
##              Total liabilities Total shareholders equity              Operating Cash Flow
##              0.005              0.045              0.878
##              Capex to Depreciation              Gross Profit Growth
##              1.000              1.000
```

A.3 Sector Mean Profiles



A.4 Full Code

All code used in this document, in execution order.

```

knitr::opts_chunk$set(
  echo = FALSE, message = FALSE, warning = FALSE,
  cache = TRUE, fig.align = "center", fig.width = 7, fig.height = 4.5
)
library(data.table)
library(dplyr)
library(tidyr)
library(ggplot2)
library(corrplot)
library(caret)          # nearZeroVar
library(factoextra)     # fviz_cluster, fviz_eig
library(cluster)
library(protoclust)     # minimax linkage
library(psych)          # fa.parallel, factor.congruence
library(lavaan)         # CFA
library(semPlot)
library(Hotelling)
library(knitr)

set.seed(442)
data_path <- "/Users/arthurjohnson/Downloads/archive/2014_Financial_Data.csv" # repo-relative
if (!file.exists(data_path)) {
  data_path <- "/Users/arthurjohnson/Downloads/archive/2014_Financial_Data.csv"
}
df <- fread(data_path)
setnames(df, "V1", "Company")

# Drop rows that are entirely missing
df <- df[!apply(is.na(df), 1, all), ]
dim(df)
ggplot(df, aes(x = factor(Class))) +
  geom_bar(fill = "steelblue") +
  labs(title = "Class Distribution (Next-Year Price Up vs. Down)",
       x = "Class", y = "Count") +
  theme_minimal()

ggplot(df, aes(x = reorder(Sector, Sector, function(x) -length(x)))) +
  geom_bar(fill = "coral") +
  labs(title = "Sector Distribution", x = "Sector", y = "Count") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
gain_threshold <- 500
n_removed <- sum(df$`2015 PRICE VAR [%]` > gain_threshold, na.rm = TRUE)
df <- df %>% filter(!(`2015 PRICE VAR [%]` > gain_threshold))
cat(sprintf("Removed %d firms with >%d%% next-year price increases\n",
           n_removed, gain_threshold))
num_cols <- names(df)[sapply(df, is.numeric)]
total_rows <- nrow(df)
missing_percent <- colSums(is.na(df[, ..num_cols])) / total_rows * 100
zero_percent <- colSums(df[, ..num_cols] == 0, na.rm = TRUE) / total_rows * 100

miss_top <- tibble(Column = names(missing_percent),
                  Percent = missing_percent) %>%

```

```

arrange(desc(Percent)) %>% head(30)

ggplot(miss_top, aes(x = reorder(Column, Percent), y = Percent)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "Variable NA-Dominance [%] (Top 30)",
       x = NULL, y = "NA-Dominance [%]") +
  theme_minimal(base_size = 10)
na_cut <- quantile(missing_percent, 0.50) # ~5.9%
zero_cut <- quantile(zero_percent, 0.60) # ~7.5%

drop_cols <- union(names(missing_percent[missing_percent > na_cut]),
                  names(zero_percent[zero_percent > zero_cut]))

# Protect identifiers and targets: Class is binary, so its zero share is
# ~50% by construction and would otherwise be dropped by the zero filter
protect <- c("Company", "Sector", "Class", "2015 PRICE VAR [%]")
drop_cols <- setdiff(drop_cols, protect)

df <- df %>% select(-all_of(drop_cols))

cat(sprintf("NA threshold: %.1f%% | Zero threshold: %.1f%%\n", na_cut, zero_cut),
    sprintf("Columns dropped: %d | Columns remaining: %d\n",
            length(drop_cols), ncol(df)))
impute_sector_revenue <- function(data, cols, threshold = 0.10) {
  data <- as.data.frame(data)
  for (col in cols) {
    miss_idx <- which(is.na(data[[col]]))
    for (i in miss_idx) {
      in_sector <- data$Sector == data$Sector[i] & !is.na(data[[col]])
      rev_i <- data$Revenue[i]
      if (!is.na(rev_i) && rev_i > 0) {
        near <- in_sector &
          abs(data$Revenue - rev_i) / rev_i <= threshold &
          !is.na(data$Revenue)
        if (any(near)) {
          data[[col]][i] <- mean(data[[col]][near], na.rm = TRUE)
          next
        }
      }
      data[[col]][i] <- median(data[[col]][in_sector], na.rm = TRUE)
    }
  }
  data
}

cols_to_impute <- names(df)[colSums(is.na(df)) > 0]
cols_to_impute <- setdiff(cols_to_impute, c("Company", "Sector"))
df <- impute_sector_revenue(df, cols_to_impute)

cat(sprintf("Remaining NAs after imputation: %d\n", sum(is.na(df))))
num_df <- df %>% select(where(is.numeric))
cor_matrix <- cor(num_df, use = "pairwise.complete.obs")

```

```

corrplot(cor_matrix, method = "color", type = "upper",
         col = colorRampPalette(c("blue", "white", "red"))(200),
         tl.pos = "n",
         title = "Correlation Heatmap of Filtered Dataset",
         mar = c(0, 0, 2, 0))
selected_vars <- c(
  "Revenue", "Gross Profit", "Operating Income", "Net Income", "EPS",
  "EBITDA", "EBIT Margin", "Profit Margin", "Free Cash Flow",
  "returnOnEquity", "Price to Sales Ratio", "Total assets",
  "Total liabilities", "Total shareholders equity", "Operating Cash Flow",
  "Capex to Depreciation", "Gross Profit Growth", "Net Income Growth",
  "2015 PRICE VAR [%]"
)
key_df <- df %>% select(any_of(selected_vars))
dim(key_df)
quantitative_vars <- df %>%
  select(where(is.numeric)) %>%
  select(-any_of(c("2015 PRICE VAR [%]", "Class")))

X <- scale(quantitative_vars)
nzv <- nearZeroVar(X, saveMetrics = TRUE)
X <- X[, !nzv$nzv]

pca_result <- prcomp(X, center = TRUE, scale. = TRUE)

variance_explained <- pca_result$sdev^2 / sum(pca_result$sdev^2)
cumulative_variance <- cumsum(variance_explained)

kable(data.frame(
  PC = 1:10,
  Proportion = round(variance_explained[1:10], 3),
  Cumulative = round(cumulative_variance[1:10], 3)
), caption = "Variance explained by the first ten principal components.")
screeplot(pca_result, type = "lines", main = "Scree Plot")
plot(cumulative_variance, type = "b",
     xlab = "Principal Component",
     ylab = "Cumulative Proportion of Variance Explained",
     main = "Cumulative Variance Explained by PCA")
abline(h = 0.80, col = "red", lty = 2)
for (i in 1:8) {
  barplot(pca_result$rotation[, i], main = paste("PC", i, "Loadings"),
         las = 2, cex.names = 0.4)
}
pc_data <- pca_result$x[, 1:8]
dist_matrix <- dist(pc_data, method = "euclidean")

hc_ward <- hclust(dist_matrix, method = "ward.D2")
plot(hc_ward, labels = FALSE, main = "Ward Hierarchical Clustering Dendrogram",
     xlab = "", sub = "")

clusters <- cutree(hc_ward, k = 6)
df$ClusterPCA <- factor(clusters)
table(df$ClusterPCA)

```

```

fviz_cluster(list(data = pc_data, cluster = clusters),
              geom = "point", ellipse.type = "convex",
              main = "Clusters in PCA Space (PC1 vs. PC2)")
hc_minimax <- protoclust(dist_matrix)
cut_mm <- protocut(hc_minimax, k = 6)
prototype_firms <- df$Company[cut_mm$protos]
kable(data.frame(Cluster = 1:6, Prototype = prototype_firms),
      caption = "Minimax linkage prototype firms for each cluster.")
ggplot(df, aes(x = ClusterPCA, y = Revenue, fill = ClusterPCA)) +
  geom_boxplot(outlier.size = 0.5) +
  scale_y_log10() +
  labs(title = "Revenue by Cluster (log scale)",
       x = "Cluster", y = "Revenue") +
  theme_minimal() + theme(legend.position = "none")

ggplot(df, aes(x = ClusterPCA, y = `2015 PRICE VAR [%]`,
              fill = ClusterPCA)) +
  geom_boxplot(outlier.size = 0.5) +
  coord_cartesian(ylim = c(-100, 200)) +
  labs(title = "Next-Year Price Change by Cluster",
       x = "Cluster", y = "2015 Price Variation [%]") +
  theme_minimal() + theme(legend.position = "none")
# The forward return is the outcome; it does not belong inside the
# factor model of the predictors
scaled_key <- scale(key_df %>% select(-any_of("2015 PRICE VAR [%]")))
parallel_result <- fa.parallel(scaled_key, fa = "fa", n.iter = 100,
                              show.legend = TRUE)
eigenvalues <- eigen(cor(scaled_key))$values
plot(eigenvalues, type = "b", main = "Scree Plot (Key Variables)",
     xlab = "Factor Number", ylab = "Eigenvalue", pch = 20, col = "blue")
abline(h = 1, col = "red", lty = 2)
n_factors <- 4
efa_result <- factanal(scaled_key, factors = n_factors, rotation = "varimax")
print(efa_result$loadings, cutoff = 0.3)
# fa.diagram() can fail on stats::factanal objects under recent psych
# versions; passing the loadings matrix directly is version-safe
fa.diagram(unclass(efa_result$loadings), main = "EFA Factor Structure")
cfa_dat <- as.data.frame(scaled_key)
colnames(cfa_dat) <- gsub(" ", "_", colnames(cfa_dat))
colnames(cfa_dat) <- gsub("\\[|\\]|%", "", colnames(cfa_dat))

model <- "
  Operational_Scale ~ Revenue + Gross_Profit + Operating_Income
  Financial_Stability ~ Total_assets + Net_Income
"

fit <- cfa(model, data = cfa_dat)
summary(fit, fit.measures = TRUE, standardized = TRUE)
semPaths(fit, what = "std", whatLabels = "std", layout = "circle",
        style = "ram", edge.label.cex = 0.8, residuals = TRUE,
        intercepts = FALSE, optimizeLatRes = TRUE)
mnv_dat <- df
colnames(mnv_dat) <- gsub(" ", "_", colnames(mnv_dat))

```

```

manova_model <- manova(
  cbind(Revenue, Net_Income, Total_assets, Gross_Profit, EBITDA,
        Gross_Profit_Growth, Operating_Cash_Flow, EPS, EBIT_Margin,
        Profit_Margin, returnOnEquity, Capex_to_Depreciation,
        Price_to_Sales_Ratio) ~ Sector,
  data = mnv_dat
)
summary(manova_model, test = "Wilks")
summary(aov(manova_model))
mnv_dat <- mnv_dat %>%
  mutate(Indicator = ifelse(`2015_PRICE_VAR_[]` > 0, 1, 0))

test_vars <- c("Revenue", "Net_Income", "Total_assets", "Gross_Profit",
               "EBITDA", "Gross_Profit_Growth", "Operating_Cash_Flow",
               "EPS", "EBIT_Margin", "Profit_Margin", "returnOnEquity",
               "Capex_to_Depreciation", "Price_to_Sales_Ratio")

group_1 <- mnv_dat %>% filter(Indicator == 1) %>% select(all_of(test_vars))
group_0 <- mnv_dat %>% filter(Indicator == 0) %>% select(all_of(test_vars))

# The raw covariance mixes dollar-level and ratio variables (variances
# spanning ~20 orders of magnitude) and is nearly collinear, so solve()
# fails on the pooled matrix. T^2 is invariant to per-variable rescaling,
# so we standardize using the pooled mean and SD, and add shrinkage
# estimation of the covariance as insurance against near-singularity.
all_X <- rbind(as.matrix(group_1), as.matrix(group_0))
ctr <- colMeans(all_X)
scl <- apply(all_X, 2, sd)
group_1 <- scale(as.matrix(group_1), center = ctr, scale = scl)
group_0 <- scale(as.matrix(group_0), center = ctr, scale = scl)

hotelling_result <- hotelling.test(group_1, group_0, shrinkage = TRUE)
print(hotelling_result)
simulate_factor_recovery <- function(n = 1000, noise_sd = 0.5, reps = 20) {
  # True loading matrix: 4 factors, 4 variables each, loading 0.8
  Lambda <- matrix(0, nrow = 16, ncol = 4)
  for (k in 1:4) Lambda[((k - 1) * 4 + 1):(k * 4), k] <- 0.8

  congruences <- replicate(reps, {
    f <- matrix(rnorm(n * 4), n, 4)
    eps <- matrix(rnorm(n * 16, sd = noise_sd), n, 16)
    Xs <- f %*% t(Lambda) + eps
    fit <- tryCatch(
      factanal(Xs, factors = 4, rotation = "varimax"),
      error = function(e) NULL
    )
    if (is.null(fit)) return(NA)
    cong <- factor.congruence(unclass(fit$loadings), Lambda)
    # Best matching between estimated and true factors
    mean(apply(abs(cong), 2, max))
  })
  mean(congruences, na.rm = TRUE)
}

```

```

noise_grid <- c(0.25, 0.5, 0.75, 1, 1.5, 2, 3)
recovery <- sapply(noise_grid, function(s)
  simulate_factor_recovery(noise_sd = s))

plot(noise_grid, recovery, type = "b", pch = 19,
  xlab = "Idiosyncratic Noise SD",
  ylab = "Mean Tucker Congruence",
  main = "Simulation 1: EFA Recovery of a Known 4-Factor Structure")
abline(h = 0.95, col = "red", lty = 2)
abline(h = 0.85, col = "orange", lty = 2)
simulate_pca_tails <- function(n = 800, p = 20, df_t = 5, reps = 20) {
  # True subspace: first two coordinate directions with sd 3 and 2
  recovery <- replicate(reps, {
    signal <- cbind(rnorm(n, sd = 3), rnorm(n, sd = 2))
    B <- qr.Q(qr(matrix(rnorm(p * 2), p, 2))) # random orthonormal basis
    noise <- matrix(rt(n * p, df = df_t), n, p)
    Xs <- signal %*% t(B) + noise
    est <- prcomp(Xs, center = TRUE, scale. = FALSE)$rotation[, 1:2]
    # Subspace similarity: mean squared canonical correlation
    mean(svd(t(est) %*% B)$d^2)
  })
  mean(recovery)
}

df_grid <- c(50, 20, 10, 5, 4, 3, 2.5)
subspace_recovery <- sapply(df_grid, function(d)
  simulate_pca_tails(df_t = d))

plot(df_grid, subspace_recovery, type = "b", pch = 19, log = "x",
  xlab = "Noise Degrees of Freedom (t-distribution, log scale)",
  ylab = "Mean Subspace Recovery",
  main = "Simulation 2: PCA Subspace Recovery Under Heavy Tails")
print(efa_result$loadings, cutoff = 0)
round(efa_result$uniquenesses, 3)
sector_means <- mnv_dat %>%
  group_by(Sector) %>%
  dplyr::summarise(across(all_of(test_vars), ~ mean(.x, na.rm = TRUE)))

sector_means_long <- pivot_longer(sector_means, cols = -Sector,
  names_to = "Variable",
  values_to = "Value")

ggplot(sector_means_long, aes(x = Variable, y = Value,
  group = Sector, color = Sector)) +
  geom_line(linewidth = 0.7) +
  geom_point(size = 2) +
  theme_minimal() +
  labs(title = "Mean Financial Metrics by Sector",
    x = "Financial Metric", y = "Mean Value") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
sessionInfo()

```

A.5 Session Info

```
## R version 4.4.1 (2024-06-14)
## Platform: aarch64-apple-darwin20
## Running under: macOS Ventura 13.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] knitr_1.48      Hotelling_1.0-8   corpcor_1.6.10   semPlot_1.1.6
## [5] lavaan_0.6-19   psych_2.4.6.26    protoclust_1.6.4  cluster_2.1.6
## [9] factoextra_1.0.7 caret_6.0-94      lattice_0.22-6    corrplot_0.95
## [13] ggplot2_4.0.2   tidyr_1.3.1       dplyr_1.2.0       data.table_1.15.4
##
## loaded via a namespace (and not attached):
## [1] RColorBrewer_1.1-3  rstudioapi_0.16.0  magrittr_2.0.3
## [4] farver_2.1.2        nloptr_2.1.1       rmarkdown_2.27
## [7] vctrs_0.7.1         minqa_1.2.8        base64enc_0.1-3
## [10] rstatix_0.7.2       htmltools_0.5.8.1  broom_1.0.6
## [13] Formula_1.2-5       pROC_1.18.5        parallelly_1.37.1
## [16] htmlwidgets_1.6.4   plyr_1.8.9         lubridate_1.9.3
## [19] igraph_2.0.3        lifecycle_1.0.5    iterators_1.0.14
## [22] pkgconfig_2.0.3     Matrix_1.7-0       R6_2.5.1
## [25] fastmap_1.2.0       future_1.33.2      digest_0.6.36
## [28] OpenMx_2.21.13      fdrtool_1.2.18     colorspace_2.1-0
## [31] Hmisc_5.2-1         ggpubr_0.6.0       labeling_0.4.3
## [34] fansi_1.0.6         timechange_0.3.0   abind_1.4-8
## [37] compiler_4.4.1      withr_3.0.0        glasso_1.11
## [40] htmlTable_2.4.3     S7_0.2.0           backports_1.5.0
## [43] carData_3.0-5       ggsignif_0.6.4     MASS_7.3-60.2
## [46] lava_1.8.0          gtools_3.9.5       ModelMetrics_1.2.2.2
## [49] tools_4.4.1         pbivnorm_0.6.0     foreign_0.8-86
## [52] zip_2.3.1           future.apply_1.11.2 nnet_7.3-19
## [55] glue_1.8.0          quadprog_1.5-8     nlme_3.1-164
## [58] lisrelTor_0.3       grid_4.4.1         checkmate_2.3.2
## [61] reshape2_1.4.5      generics_0.1.3     recipes_1.1.0
## [64] gtable_0.3.6        class_7.3-22       car_3.1-3
## [67] utf8_1.2.4          sem_3.1-16         ggrepel_0.9.6
## [70] foreach_1.5.2       pillar_1.9.0       stringr_1.5.1
## [73] rockchalk_1.8.157   splines_4.4.1      survival_3.6-4
## [76] kutils_1.73         tidyselect_1.2.1   pbapply_1.7-2
## [79] gridExtra_2.3       stats4_4.4.1       xfun_0.45
## [82] hardhat_1.4.0       qgraph_1.9.8       timeDate_4041.110
```

## [85]	arm_1.14-4	stringi_1.8.4	yaml_2.3.9
## [88]	boot_1.3-30	evaluate_0.24.0	codetools_0.2-20
## [91]	mi_1.1	tibble_3.2.1	cli_3.6.3
## [94]	RcppParallel_5.1.9	rpart_4.1.23	xtable_1.8-4
## [97]	Rcpp_1.0.12	globals_0.16.3	coda_0.19-4.1
## [100]	png_0.1-8	XML_3.99-0.17	parallel_4.4.1
## [103]	gower_1.0.1	jpeg_0.1-10	lme4_1.1-35.5
## [106]	listenv_0.9.1	ipred_0.9-14	scales_1.4.0
## [109]	prodlim_2024.06.25	openxlsx_4.2.7.1	purrr_1.0.2
## [112]	rlang_1.1.7	mnormt_2.1.1	